

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2024

TEST DATE: 14-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

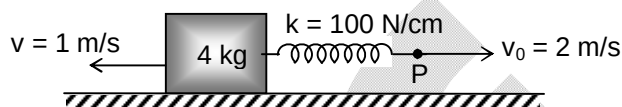
Physics

PART – A

SECTION – A (One Options Correct Type)

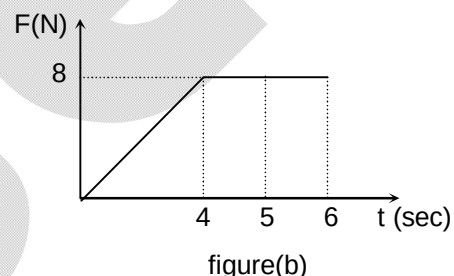
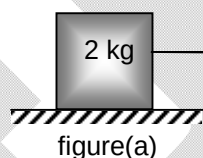
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A spring block system lies on a smooth horizontal surface. The free end of the spring is being pulled towards right with constant speed $v_0 = 2 \text{ m/s}$. At $t = 0$, the spring (spring constant $k = 100 \text{ N/cm}$) is unstretched and block is having a speed $v = 1 \text{ m/s}$ to the left. The maximum acceleration of the block during the motion is
- (A) 50 m/s^2
(C) 150 m/s^2



- (B) 100 m/s^2
(D) 200 m/s^2

2. A block of mass 2 kg is kept on a rough horizontal surface having coefficient of friction ($\mu_s = 0.3$ and $\mu_k = 0.25$). A time dependent force F is applied on the block as shown in the figure (b). Find the speed of the block at $t = 5 \text{ sec}$.



- (A) 12 m/s
(C) 2.5 m/s

- (B) zero
(D) 0.5 m/s

3. A container of fixed volume has mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T . Assuming the gases are ideal, then choose the INCORRECT Statement.
- (A) Internal energy of the mixture is $4RT$ Joule

- (B) Ratio of rms speed of gas molecules in the mixture and hydrogen is $\sqrt{\frac{2}{3}}$

- (C) Ratio of speed of sound in the mixture and helium is $\sqrt{\frac{6}{5}}$

- (D) Ratio of rms speed of the gas molecules in Helium and hydrogen is $\sqrt{2}$

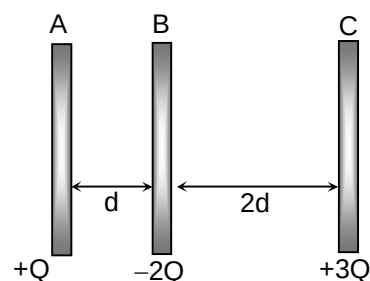
4. Three large metallic parallel plates A, B and C each having area A are separated by very small distances as shown in the figure. Charges Q , $-2Q$ and $3Q$ are given to the plates. The potential difference between plates C and A is

- (A) zero

- (B) $\frac{2Qd}{\epsilon_0 A}$

- (C) $\frac{4Qd}{\epsilon_0 A}$

- (D) Indeterminate

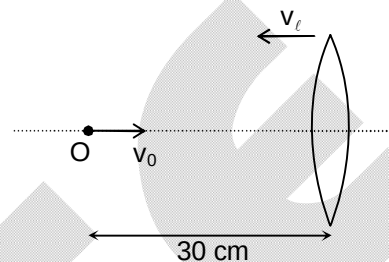


5. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$ and comes out without any change in magnitude or direction of $\vec{v}$. Then (here $\vec{E}$ is electric field and $\vec{B}$ is magnetic field)

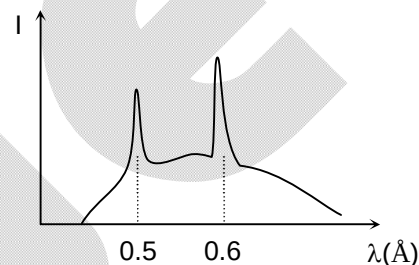
(A) $\vec{B} \times \vec{E} = |\vec{E}|^2 \vec{v}$ (B) $\vec{E} \times \vec{B} = |\vec{B}|^2 \vec{v}$
 (C) $\vec{B} \times \vec{E} = |\vec{B}|^2 \vec{v}$ (D) $\vec{E} \times \vec{B} = |\vec{E}|^2 \vec{v}$

6. A point object has velocity $v_0 = 2$ cm/s when it is placed at a distance of 30 cm from a lens moving with a speed $v_\ell = 2$ cm/s towards left. If the focal length of the lens is $f = 20$ cm, find the velocity of the image with respect to the ground.

- (A) 14 cm/s towards right
 (B) 14 cm/s towards left
 (C) 25 cm/s towards right
 (D) 25 cm/s towards left



7. The graph shows the intensity of X-rays emitted by a heavy target and the wavelength, when electrons of energy 20 keV are incident on the target. In the graph one peak is of K_α line and the other peak is of K_β line. Then



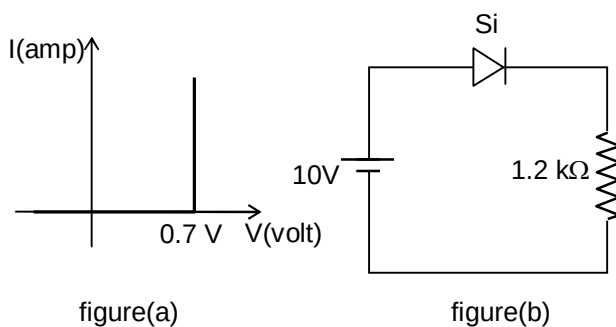
- (A) If the energy of incident electron is increased to 30 keV, then both the peaks will shift rightward.
 (B) If the energy of incident electron is increased to 30 keV, then both the peaks will shift leftward.
 (C) K_α line wavelength is $0.5 \text{ \AA}$
 (D) K_α line wavelength is $0.6 \text{ \AA}$

8. A 2 kg particle is moving along the x-axis under the action of the force $\vec{F} = -\frac{\pi^2}{32} \vec{x}$, where F is in newton and x is in meter. At $t = 2$ sec, the particle is moving along $\vec{x}$ and is at $x = 0$. At $t = 10$ sec it is observed that particle's speed is $4\sqrt{2}$ m/s. Then amplitude of particle motion is

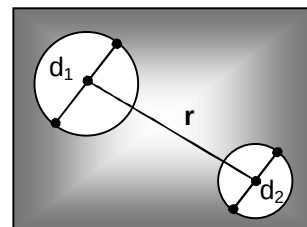
(A) $\frac{32}{\pi} \text{ m}$ (B) $\frac{32\sqrt{2}}{\pi} \text{ m}$
 (C) $\frac{16\sqrt{2}}{\pi} \text{ m}$ (D) $\frac{16}{\pi} \text{ m}$

9. In the figure (a) I-V characteristic of Si-diode is shown. In figure(b) shows a circuit having the Si-diode. The current in the circuit is

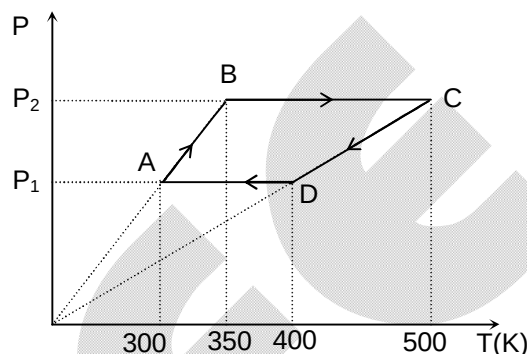
- (A) 8.33 mA
 (B) 0
 (C) 8.92 mA
 (D) 7.75 mA



10. Two holes of unequal diameters d_1 and d_2 ($d_2 < d_1$) cut in metal sheet. Separation between the centres of the holes is r . Now the sheet is heated to a higher temperature. Then
- (A) d_1, d_2 increases and r decreases
 (B) d_1, d_2 increases and r increases
 (C) d_1, d_2 decreases and r increases
 (D) d_1, d_2 decreases and r decreases



11. P-T curve of a cyclic process is shown. If number of moles of the monoatomic ideal gas are n . Then efficiency of the cycle is
- (A) 1
 (B) $1/3$
 (C) $1/9$
 (D) $1/11$

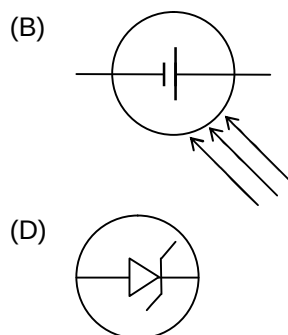
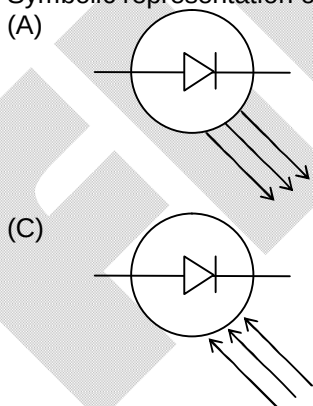


12. In an xyz space only electric field is present having potential function $V = k(x^2 + y^2 + z^2)$ volt, where k is a positive constant and x, y, z are co-ordinate of a point in meter unit. A positive charge particle of charge q and mass m is placed at $(1m, 1m, 1m)$. Acceleration of the particle at the point is

- (A) $\frac{6kq}{m}$
 (B) $\frac{\sqrt{3}kq}{m}$
 (C) $\frac{\sqrt{3}kq}{2m}$
 (D) $\frac{2\sqrt{3}kq}{m}$

13. A nucleus X of mass number 218 in free state decays to emit an α -particle. Kinetic energy of the α -particle emitted is 6.7 MeV. The Q-value of the α -decay nuclear reaction is (in MeV)
- (A) 0.125
 (B) 6.825
 (C) 6.8
 (D) 6.7

14. Symbolic representation of photodiode is

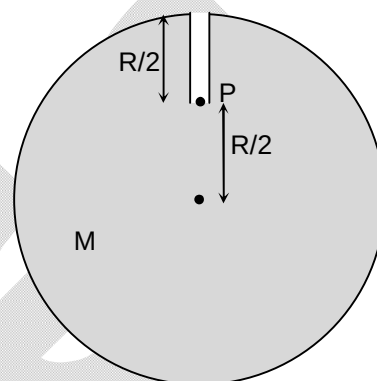


15. Two coherent sound sources in phase of frequencies of 256 Hz and 260 Hz are sounding together. If an observer records an intensity of I_0 at $t = 0$, then he will observe intensity at $t = \frac{1}{16}$ sec.

(A) I_0 (B) $\frac{I_0}{2}$
(C) $\frac{I_0}{4}$ (D) zero

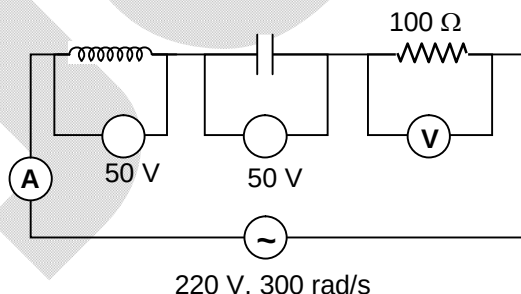
16. A particle is projected from a depth $\frac{R}{2}$ inside the earth such that it escapes to infinity. Mass of the particle is negligibly small is compared to mass of earth M . Then escape speed from the point P is

(A) $\sqrt{\frac{2GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$
(C) $\sqrt{\frac{13GM}{4R}}$ (D) $\sqrt{\frac{11GM}{4R}}$



17. For the AC circuit in the figure, choose the correct option having reading of AC voltmeter (V) and AC ammeter (A).

(A) 120 V, 1.2 A
(B) 220 V, 2.2 A
(C) 320 V, 3.2 A
(D) 220V, 1.2 A



18. The color code of a resistance is given in the figure. The value of resistance and tolerance, respectively are

(A) 47 k Ω , 10%
(B) 4.7 k Ω , 5%
(C) 470 Ω , 5%
(D) 470 k Ω , 5%

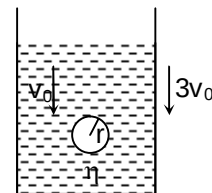


19. A non conducting ring of mass m and radius R a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that the plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0 t^2$ tesla is switched on at $t = 0$. After 2 second for switching on the magnetic field the ring is just about to rotate about vertical axis through its centre. m , R , Q , B are in SI unit. Then

(A) The induced electric field is steady with respect to time in the space
(B) the net electric force acting on the ring at any time t is $QB R t$
(C) Friction first acts at $t = 2$ sec

(D) The coefficient of static friction between the ring and the surface is $\frac{2B_0 R Q}{mg}$

20. A container filled with viscous liquid is moving vertically downward with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertical downwards in liquid has a speed v_0 with respect to ground. The coefficient of viscosity is η . There is no relative motion between the container and the liquid. Then at the shown instant, the power developed by viscous force with respect to ground frame is



- (A) $-6\pi\eta rv_0^2$ (B) zero
(C) $12\pi\eta rv_0^2$ (D) $-12\pi\eta rv_0^2$

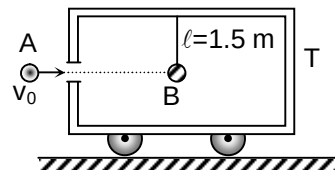
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

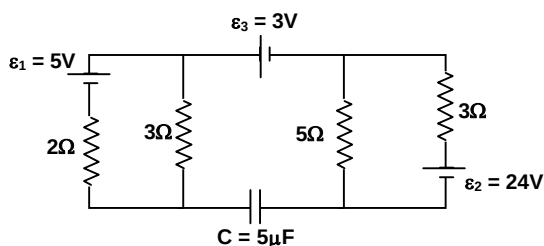
21. An unpolarized light of intensity 80 W/m^2 passes through three polarizing sheets whose pass axes make an angle of 30° , 60° and 30° with y-axis in same sense. The final transmitted intensity after passing through all three polarizers is $I \text{ W/m}^2$. Then find the value of $2I$.
22. A particle of specific charge $2 \times 10^{-2} \text{ C/kg}$ starts moving from the origin under the action of a uniform electric field $\vec{E} = (100 \text{ N/C})\hat{i}$ and magnetic field $\vec{B} = (5\text{T})\hat{k}$. Its velocity vector at $(a, b, 0) \text{ m}$ is $(4\hat{i} - 3\hat{j}) \text{ m/s}$. Then value of $4a$ is (in m)
23. An object is projected with a velocity of 10 m/s making an angle of 30° with horizontal from origin of x-y vertical plane having positive x-axis along horizontal and positive y-axis along the vertically upward direction. The equation of the trajectory is $y = Ax - Bx^2$ where x and y are co-ordinates at a point in meter. Find the value of $\sqrt{3} \left(\frac{A}{B} \right)$ (in SI unit). (Take $g = 10 \text{ m/s}^2$)
24. Each side of a cube is measured to be 7.203 m . If the total surface area of the cube considering the correct significant digits in m^2 is A , then find the value of $10A$.
25. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in 40 seconds is $\tau \text{ N-m}$. Find the value of $\pi\tau$. (Take $\pi^2 = 10$)

26. In the figure shown a ball A of mass $m = 1 \text{ kg}$ collide with another identical ball B with a speed $v_0 = 6 \text{ m/s}$ and sticks to it. The ball B is hanging inside a trolley T of mass $M = 4 \text{ kg}$. Find the maximum height (in cm) attained by the ball (A + B) from its initial level after the collision (Take $g = 10 \text{ m/s}^2$)



27. The solar constant (intensity of solar radiation) just at Venus surface measured to be 2600 W/m^2 . If the radius of the Sun is $7 \times 10^8 \text{ m}$, while the Venus-Sun distance is $1.1 \times 10^{11} \text{ m}$, then find the most probable wavelength in the solar radiation in nm. [Take $(1.81)^{\frac{1}{4}} = 1.16$, wein's constant $b = 2.9 \times 10^{-3} \text{ m-K}$] Sun can be assumed as a black body.

28. If energy stored in the capacitor C in the steady state is $U_0 \mu\text{J}$, then find $2U_0$.



29. A parallel plate capacitor with $A = 1.2 \text{ m}^2$ and separation $d = 1 \text{ mm}$ is charged by a current $i = (8\text{A})e^{-\frac{t}{4\text{sec}}}$. Consider a plane surface X of area 0.6 m^2 parallel to the plates and drawn symmetrically between the plates. Find the displacement current (in mA) through the plane surface X at $t = 4 \text{ sec}$. (Take $e^{-1} = 0.37$)
30. A submarine A travelling at 18 km/hr is being chased along the line of its velocity by another submarine B travelling at 27 km/hr . B send a Sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν (in Hz) is close to (take speed of sound in water = 1500 m/s)

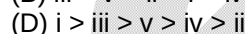
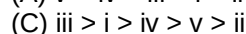
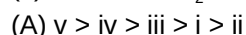
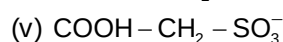
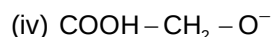
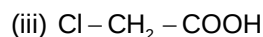
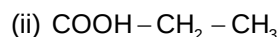
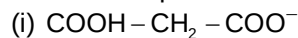
Chemistry

PART – B

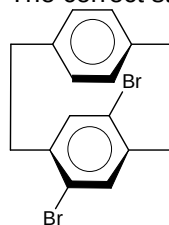
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Compare the acidic strength of following compounds and arrange in decreasing order. Choose the correct option



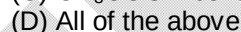
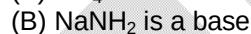
32. The correct statement about the following molecule is:



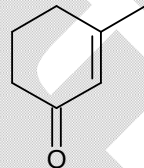
- (A) Molecule is chiral and does not possess chiral centre.
 (B) Molecule is chiral and possess chiral centre.
 (C) Molecule is achiral due to plane of symmetry.
 (D) Molecule is achiral due to axis of symmetry.

33. Choose the correct statement:

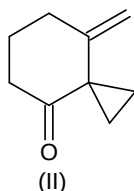
In liquid ammonia:



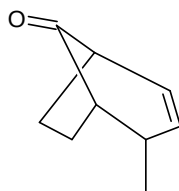
34. The product(s) are :



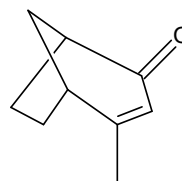
(I)



(II)



(III)



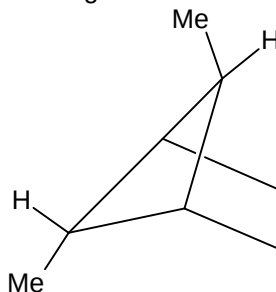
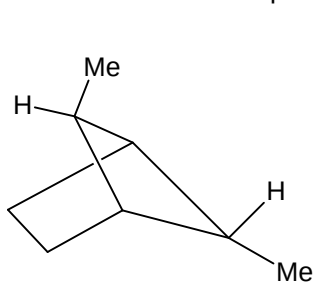
(IV)

- (A) I and III
 (C) I and II

- (B) II and IV
 (D) II and III

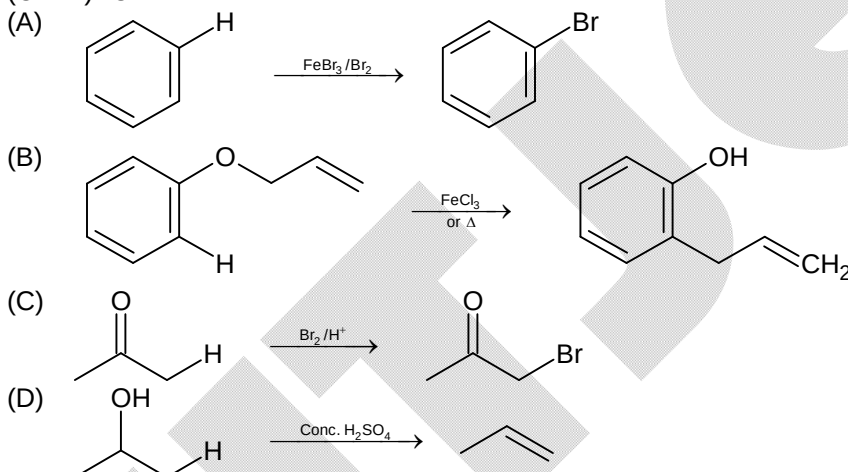
35. Considering N as central atom, the structure of $(\text{CH}_3)_2\text{NCS}$ and $(\text{SiH}_3)_2\text{NCS}$ respectively, are:
 (A) Linear and Bent (B) Bent and Bent
 (C) Linear and Linear (D) Bent and linear

36. The correct relationship between the following structures is that they are:



- (A) Enantiomers (B) Identical
 (C) Diastereomers (D) Constitutional isomers

37. The reaction that is expected to show primary kinetic isotope effect for the indicated H-atom (C – H) is:



38. The pair in which both actinides show +3 oxidation state only is:

- (A) Ac and Lr (B) Ac and No
 (C) Cm and Bk (D) Cm and Lr

39. I_2 is violet in the solid as well as in gas phase. However, in acetone or ethanol, it turns brown. Choose the correct statement(s) for this colour change:

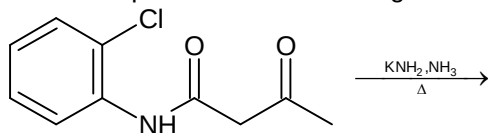
- (i) Dissociation of I_2 in atomic state.
 (ii) Interaction of low-lying σ^* orbital of iodine with lone pair of O (solvent).
 (iii) Formation of a charge transfer complex.

- (A) (i) only (B) (ii) only
 (C) (i) and (ii) only (D) (iii) only

40. An atomic orbital is given by $nr^2 e^{-\frac{r}{3a_0}} (3\cos^2\theta - 1)$ in hydrogen atom. The orbital represents:

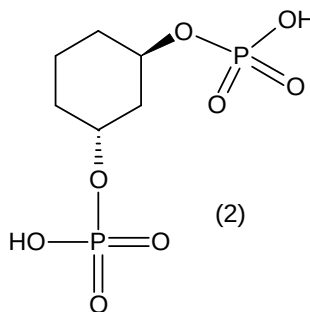
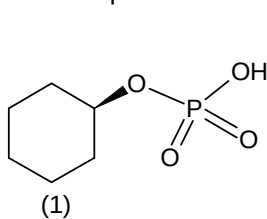
- (A) 2p (B) 3p
 (C) 3d (D) 4d

41. The magnitude of bond angles in gaseous NF_3 , SbF_3 and SbCl_3 follows the order:
 (A) $\text{NF}_3 > \text{NH}_3 > \text{SbCl}_3$ (B) $\text{SbCl}_3 > \text{NH}_3 > \text{NF}_3$
 (C) $\text{NH}_3 > \text{NF}_3 > \text{SbCl}_3$ (D) $\text{NF}_3 > \text{SbCl}_3 > \text{NH}_3$
42. Consider the following chemical reactions in gas phase
 Reaction I : $\text{H} + \text{D}_2 \longrightarrow \text{HD} + \text{D}$
 Reaction II : $\text{D} + \text{H}_2 \longrightarrow \text{HD} + \text{H}$
 Which of the following statement is true?
 (A) Reaction I and II, both are endothermic.
 (B) Reaction I is endothermic, Reaction II is exothermic.
 (C) Both reactions are thermo neutral as they involve isotopes which differ by the number of neutrons, which do not play any role in chemical reactions.
 (D) Reaction I and II both are exothermic.
43. Predict the product of the following reaction

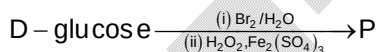


- (A)
- (B)
- (C)
- (D)

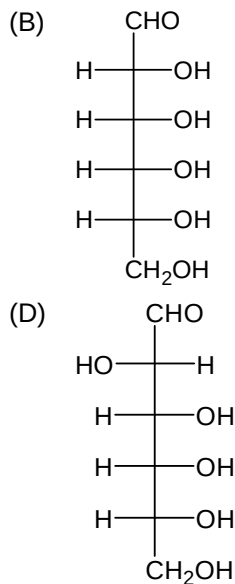
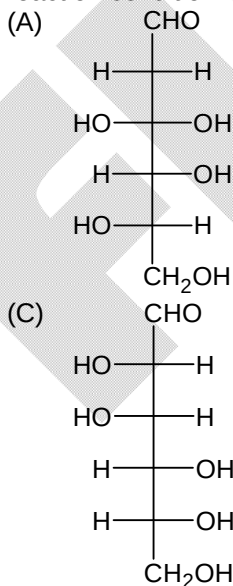
44. Compound 2 and compound 1 were mixed in a 1 : 4 molar ratio. The mixture was subjected to an ammonium molybdate test for phosphate estimation. The result of the molybdated test yielded a phosphate concentration of 36.0 mM for the mixture. What are the concentration of compound 1 and compound 2 in the mixture?



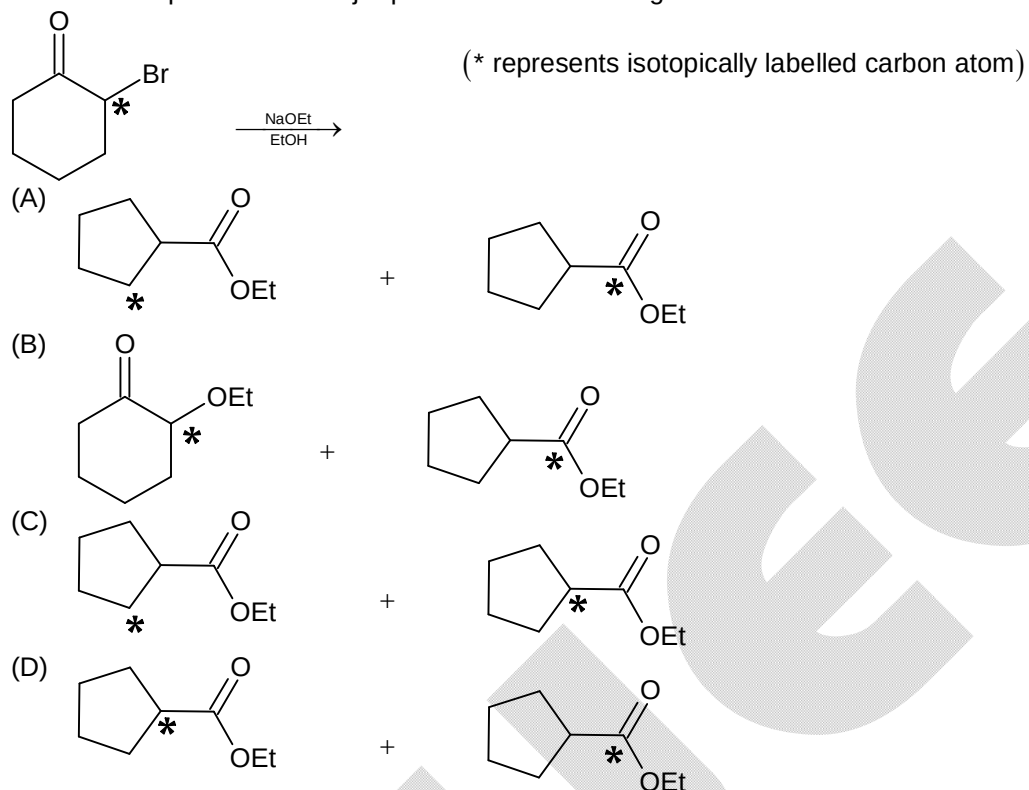
- (A) Compound 1 : 30 mM Compound 2 : 6 mM
 (B) Compound 1 : 24 mM Compound 2 : 6 mM
 (C) Compound 1 : 28.8 mM Compound 2 : 7.2 mM
 (D) Compound 1 : 24 mM Compound 2 : 12 mM
45. Which of the following can't be used for the given transformation?
- $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{C}(\text{CH}_3)_3 \longrightarrow \text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)_3$
- (A) Clemmenson reduction (B) Wolf-Kishner reduction
 (C) Mozingo reduction (D) Bouveault-Blanc reduction
46. A mixture of acetone and CCl_4 can be separated by
- (A) Azeotropic distillation (B) Fractional distillation
 (C) Steam distillation (D) Vacuum distillation
47. In the following reaction of D-(+)glucose a product P is formed.



Amongst the following compounds, the one which will give the same product (P) under identical reaction condition is:



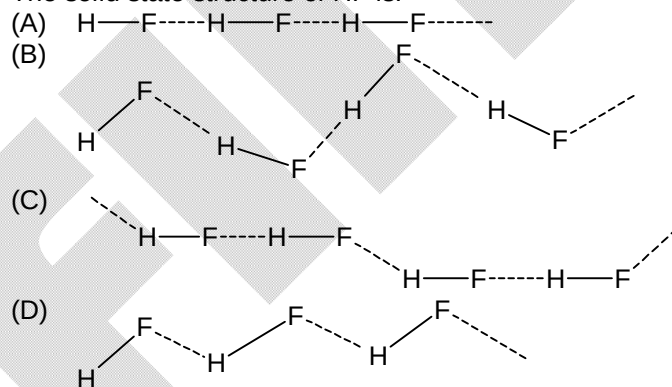
48. The correct option for the major product of the following reaction is:



49. The salt bridge in a Galvanic cell allows the flow of:

(A) ions but not electrons
 (B) both ions and electrons
 (C) electrons but not ions
 (D) neither ions nor electrons

50. The solid state structure of HF is:

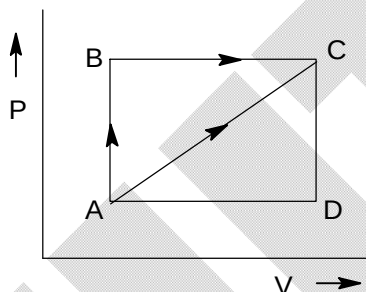


SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The number of stereoisomers for the complex $[\text{RhH}(\text{C} \equiv \text{CR})_2(\text{PMe}_3)_3]$ is:
52. The total number of lone pairs of electron on all atoms in cyanogens is 'x' and thiocyanogen is 'y'. Report $(x + y) = ?$
53. For the cell $\text{Cd} | \text{CdCl}_2 || \text{AgCl} | \text{Ag}$; $E^\circ_{\text{cell}} = 0.675 \text{ V}$ and $\left(\frac{dE^\circ}{dT}\right) = -6.5 \times 10^{-4} \text{ VK}^{-1}$ at 27°C . The $\Delta H^\circ (\text{kJ mol}^{-1})$ value for the cell is:
54. An organic compound is estimated through Dumas method and was found to evolve 264 g CO_2 , 72 g of H_2O and 28 g of N_2 . If the formula of the compound is $\text{C}_x\text{H}_y\text{N}_z$. Then $\left(\frac{x+y}{z}\right)$ is:
55. When glycinium hydrochloride is titrated against NaOH, pH at the first half equivalence point is 2.40 and the pH at the second half equivalence point is 9.60. The pH at first equivalence point is:
56. A thermodynamic process is shown below:
 Given: $P_A = 3 \times 10^4 \text{ Pa}$, $P_B = 8 \times 10^4 \text{ Pa}$, $V_A = 2 \times 10^{-3} \text{ m}^3$, $V_D = 5 \times 10^{-3} \text{ m}^3$. In the process AB, 600 J of heat is added and in BC 200 J of heat is added to the system. If the change in internal energy of the system in the process AC is x; then report $\left(\frac{x}{20}\right)$. x is calculated in Joule.



57. 10.6 g of benzaldehyde was reacted with an excess of acetophenone to produce 10.4 g of the enone product according to the given equation. Calculate the % yield of the product.
 $\text{Benzaldehyde} + \text{Acetophenone} \xrightarrow[2. \Delta]{1. \text{NaOEt}} \text{Product}$
58. Assume that the reaction of MeMgBr with ethyl acetate proceeds with 100% conversion to tert – butanol. The volume of 0.2 M solution of MeMgBr required to convert 10 mL of a 0.025 M solution of ethyl acetate is x. Report the value of $10x$ in mL.

59. The mechanism of enzyme catalysis is as follows:



The rate of product formation is given as:

$$\frac{+dP}{dt} = \frac{k_1 k_2 [E_0][S]}{k_{-1} + k_2 + k_1 [S]}, \text{ where } [E_0] \text{ is the total enzyme concentration, } [S] \text{ is substrate concentration.}$$

The rate of product formation when the substrate concentration is very large has a limiting value of 0.02 Ms^{-1} . At a substrate conc. of 250 mg/dm^3 , the rate is half this value. The value of $\frac{k_1}{k_{-1}}$ (in dm^3 / kg) assuming that $k_2 \ll k_{-1}$ is $x \times 10^3$. What is the value of x ?

60. The maximum concentration of equimolar solution of ferrous sulphate and potassium sulphide so that when mixed in equal volumes, there is no precipitate of iron sulphide is $C \times 10^{-9} \text{ M}$. Report the value of C . [Given: $K_{sp}(\text{FeS}) = 6.3 \times 10^{-18} \text{ M}^2$.

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If a non constant function $f(x) = \frac{2x^2 - kx + 5}{x^2 + 3x + 4}$ is bounded, then k can not have values is equal to
 (A) -2 (B) -6
 (C) 2 (D) 6
62. If $\sqrt{2} < \alpha, \beta < \sqrt{3}$, $\alpha \neq \beta$, then the value of $\left[\frac{\alpha - \beta}{\cot^{-1} \alpha - \cot^{-1} \beta} \right]$ is (where [.] denotes G.I.F.)
 (A) -4 (B) 4
 (C) 3 (D) -5
63. Let real number a satisfy $\{a\} + \left\{\frac{1}{a}\right\} = 1$ and b is the value of $\{a^3\} + \left\{\frac{1}{a^3}\right\}$ (where {.} denotes fractional part function), then the value of $\int_0^b \frac{x}{1+x} \sqrt{1-x^2} dx$ is
 (A) $2 + \frac{\pi}{4}$ (B) $2 - \frac{\pi}{4}$
 (C) $1 - \frac{\pi}{4}$ (D) $1 + \frac{\pi}{4}$
64. For a real number 'a' such that the equation $a(\sin x + \sin(2x)) = \sin 3x$ has more than one solution in the interval $(0, \pi)$ the set of all such value of a is
 (A) $(-3, -2)$ (B) $(-3, 1) - \{-2\}$
 (C) $(-2, 1)$ (D) $(0, 1)$
65. Let c be a real number and let z_1 and z_2 be the two complex number satisfying the equation $z^2 - cz + 10 = 0$, points $z_1, z_2, \frac{1}{z_1}$ and $\frac{1}{z_2}$ are vertices of (convex) quadrilateral in the complex plane, when the area is maximum then the value of c is
 (A) $4\sqrt{5}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{5}$ (D) $8\sqrt{5}$
66. If the equation $\frac{\alpha^2}{1 - \tan^2 x} = \frac{\sin^2 x + \alpha^2 - 2}{\cos 2x}$ has a solution, then
 (A) $\alpha^2 < 1$ (B) $\alpha^2 > 2$
 (C) $\alpha^2 \geq 1$ (D) $\alpha \in \mathbb{R}$
67. A polynomial of degree four with leading coefficient 1 and integer coefficients, has two zeros, both of which are integer. Which of the following can also be a zero of the polynomial ?
 (A) $\frac{1+i\sqrt{11}}{2}$ (B) $\frac{1+i}{2}$
 (C) $\frac{1}{2} + i$ (D) $1 + \frac{i}{2}$

68. Circle $x^2 + y^2 - 6x - 4y + 11 = 0$ is rolled up by 4 units along a tangent to it at the point (2, 3) let this be circle C_1 , C_2 is the mirror image of circle C_1 about the tangent. A and B are centres of circles C_1 and C_2 , C and D are the feet of perpendicular from A and B respectively upto x-axis, then the area of trapezium ABCD equals to
- (A) $(2 + 2\sqrt{2})$ (B) $(3 + 2\sqrt{2})$
 (C) $(6 + 4\sqrt{2})$ (D) $(4 + 4\sqrt{2})$
69. A point P is selected at random from the interior of the pentagon with vertices $A = (0, 2)$, $B = (4, 0)$, $C = (2\pi + 1, 0)$, $D = (2\pi + 1, 4)$ and $E = (0, 4)$. What is the probability that $\angle APB$ is obtuse ?
- (A) $1/5$ (B) $1/4$
 (C) $3/8$ (D) $5/16$
70. How many 15 letter arrangements of 5 A's, 5B's and 5 C's have no A's in the first 5 letter no B's in the next 5 letters and no C's in the last 5 letters
- (A) $\sum_{k=0}^5 ({}^5C_k)^3$ (B) $\sum_{k=0}^5 ({}^5P_k)^3$
 (C) 2^{15} (D) $2^5 \cdot 3^5$
71. The value of $\sum_{i=0}^n \sum_{j=1}^n {}^nC_j \cdot {}^jC_i$ $i \leq j$
- (A) 3^n (B) $3^n + 1$
 (C) $3^n - 1$ (D) 3^{n+1}
72. A parabola $y = ax^2 + bx + c$ is reflected about the x-axis, the parabola and its reflection are translated horizontally five units in opposite direction to become the graph of $y = f(x)$ and $y = g(x)$ respectively. Then which of the following describes the graph of $y = f(x) + g(x)$
- (A) A horizontal line (B) the graph of a cubic function
 (C) a non horizontal line (D) A parabola
73. If z_1 and z_2 are two complex number such that $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ where $z_1 = 4 + 6i$ and $z_2 = 3 + 5i$, then locus of z is
- (A) straight line with slope $-\frac{1}{2}$ (B) circle with radius $\frac{1}{\sqrt{2}}$
 (C) hyperbola with eccentricity $\sqrt{2}$ (D) hyperbola with eccentricity $\frac{5}{2}$
74. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$, then $\int_{1/10}^{9/10} f(f(x))dx$ is equal to
- (A) $\frac{4}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{5}$ (D) 0
75. If $I_1 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}} x^2 dx$ and $I_2 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}+1} x^2 dx$, then $\frac{I_1 - I_2}{I_2} = \frac{1}{k}$, then k is equal to
- (A) $(\sqrt{3} + 1)\sqrt{2}$ (B) $(\sqrt{3} - 1)\sqrt{2}$
 (C) $(\sqrt{3} + \sqrt{2})$ (D) $3(\sqrt{6} + \sqrt{2})$

76. Total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^{10} + 1}{t^{10} + t^4 + 1} dt = 6x - 5$ is
 (A) 0 (B) 1
 (C) 2 (D) 3
77. A continuous and differentiable function $f(x)$ satisfy the functional equation $f(x + y) = \frac{f(x)f(y)}{f(x) + f(y)}$,
 $f(1) = 32$ then $f(4) =$
 (A) 8 (B) 6
 (C) 4 (D) 2
78. $\lim_{x \rightarrow \infty} |1 - a^4|^x \sin \frac{5}{|1 - a^2|^x} a^2 > 2$ is
 (A) 0 (B) 5
 (C) ∞ (D) does not exist
79. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2 - x)$ and $f''(x) < 0 \forall x \in (0, 2)$ then which is true ?
 (A) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $(0, 2)$
 (B) $g(x)$ increases in $(0, 2)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (C) $g(x)$ increases in $\left(\frac{4}{3}, 2\right)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (D) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $\left(\frac{4}{3}, 2\right)$
80. Area bounded by the curve $y = 1$ and $y = \frac{\sin x + \cos x + |\sin x - \cos x|}{2}$ for $x \in [0, 2\pi]$ is
 (A) $2\pi - \sqrt{2}$ (B) $2\pi - 2\sqrt{2}$
 (C) $2(\pi - \sqrt{2} - 1)$ (D) $2(\pi + \sqrt{2} - 3)$

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = \sqrt{ax^2 + bx}$, for how many real values of a is there at least one positive value of b for which the domain of $f(x)$ and the range of $f(x)$ are the same set
82. Lines $x + y + 4 = 0$, $2x + ky + 2 = 0$ intersects the circle $x^2 + y^2 + 2x + 4y + 1 = 0$ and $x^2 + y^2 + 4x + 6y + 4 = 0$ as 4 points which are concyclic, then the value of k is
83. The number of distinct values of determinant of order 2×2 whose entries are from the set $\{-1, 0, 1\}$ is
84. Krishna writes the number 1, 2, 3, ..., 9 on separate cards, one number per card, he wishes to divide the cards into 3 groups of 3 cards so that the sum of the numbers in each group will be the same. In how many ways can this be done ?

85. Fifteen integers $a_1, a_2, a_3, \dots, a_{15}$ are arranged in order on a number line. The integers are equally spaced and have the property that $1 \leq a_1 \leq 10$, $13 \leq a_2 \leq 20$ and $241 \leq a_{15} \leq 250$. What is the sum of digits of a_{14} ?
86. In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term is 4000. What is the first term
87. A data set consists of 6 (not all distinct) positive integers : 1, 7, 5, 2, 5 and x, the Arithmetic Mean of the 6 numbers equals a value in the data set. What is the sum of all possible values of x
88. The area bounded by the graph of $y^2 + 2xy + 40|x| = 400$ is
89. Let the distance between two point (x_1, y_1) and (x_2, y_2) is given by $|x_1 - x_2| + |y_1 - y_2|$ for how many points P with integer coordinates is the distance between P and the origin less than or equal to 20
90. Two integers are inserted into list 3, 3, 3, 8, 11, 12, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of the two additional numbers

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2024

TEST DATE: 14-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

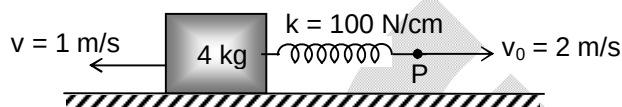
Physics

PART – A

SECTION – A (One Options Correct Type)

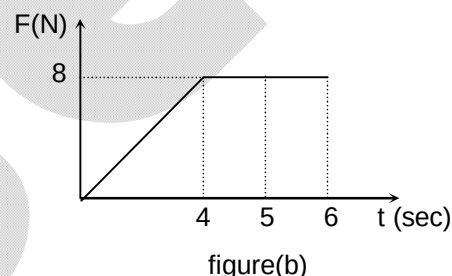
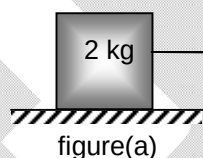
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A spring block system lies on a smooth horizontal surface. The free end of the spring is being pulled towards right with constant speed $v_0 = 2 \text{ m/s}$. At $t = 0$, the spring (spring constant $k = 100 \text{ N/cm}$) is unstretched and block is having a speed $v = 1 \text{ m/s}$ to the left. The maximum acceleration of the block during the motion is
- (A) 50 m/s^2
(C) 150 m/s^2



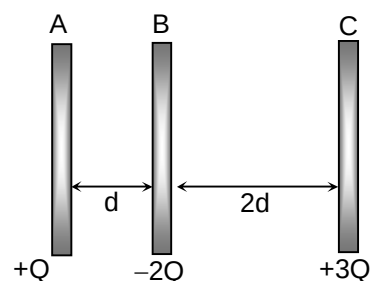
- (B) 100 m/s^2
(D) 200 m/s^2

2. A block of mass 2 kg is kept on a rough horizontal surface having coefficient of friction ($\mu_s = 0.3$ and $\mu_k = 0.25$). A time dependent force F is applied on the block as shown in the figure (b). Find the speed of the block at $t = 5 \text{ sec}$.



- (A) 12 m/s
(C) 2.5 m/s
- (B) zero
(D) 0.5 m/s
3. A container of fixed volume has mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T . Assuming the gases are ideal, then choose the INCORRECT Statement.
- (A) Internal energy of the mixture is $4RT$ Joule
(B) Ratio of rms speed of gas molecules in the mixture and hydrogen is $\sqrt{\frac{2}{3}}$
(C) Ratio of speed of sound in the mixture and helium is $\sqrt{\frac{6}{5}}$
(D) Ratio of rms speed of the gas molecules in Helium and hydrogen is $\sqrt{2}$

4. Three large metallic parallel plates A, B and C each having area A are separated by very small distances as shown in the figure. Charges Q , $-2Q$ and $3Q$ are given to the plates. The potential difference between plates C and A is
- (A) zero
(B) $\frac{2Qd}{\epsilon_0 A}$
(C) $\frac{4Qd}{\epsilon_0 A}$
(D) Indeterminate



5. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$ and comes out without any change in magnitude or direction of $\vec{v}$. Then (here $\vec{E}$ is electric field and $\vec{B}$ is magnetic field)

(A) $\vec{B} \times \vec{E} = |\vec{E}|^2 \vec{v}$

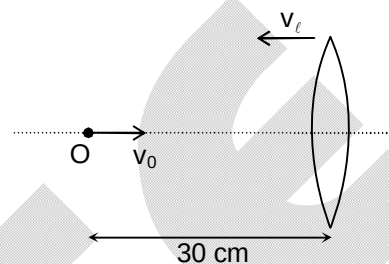
(B) $\vec{E} \times \vec{B} = |\vec{B}|^2 \vec{v}$

(C) $\vec{B} \times \vec{E} = |\vec{B}|^2 \vec{v}$

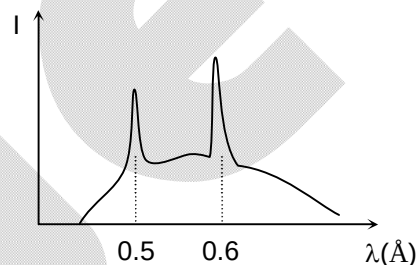
(D) $\vec{E} \times \vec{B} = |\vec{E}|^2 \vec{v}$

6. A point object has velocity $v_0 = 2$ cm/s when it is placed at a distance of 30 cm from a lens moving with a speed $v_\ell = 2$ cm/s towards left. If the focal length of the lens is $f = 20$ cm, find the velocity of the image with respect to the ground.

- (A) 14 cm/s towards right
(B) 14 cm/s towards left
(C) 25 cm/s towards right
(D) 25 cm/s towards left



7. The graph shows the intensity of X-rays emitted by a heavy target and the wavelength, when electrons of energy 20 keV are incident on the target. In the graph one peak is of K_α line and the other peak is of K_β line. Then



- (A) If the energy of incident electron is increased to 30 keV, then both the peaks will shift rightward.
(B) If the energy of incident electron is increased to 30 keV, then both the peaks will shift leftward.
(C) K_α line wavelength is $0.5 \text{ \AA}$
(D) K_α line wavelength is $0.6 \text{ \AA}$

8. A 2 kg particle is moving along the x-axis under the action of the force $\vec{F} = -\frac{\pi^2}{32} \vec{x}$, where F is in newton and x is in meter. At $t = 2$ sec, the particle is moving along $\vec{x}$ and is at $x = 0$. At $t = 10$ sec it is observed that particle's speed is $4\sqrt{2}$ m/s. Then amplitude of particle motion is

(A) $\frac{32}{\pi} \text{ m}$

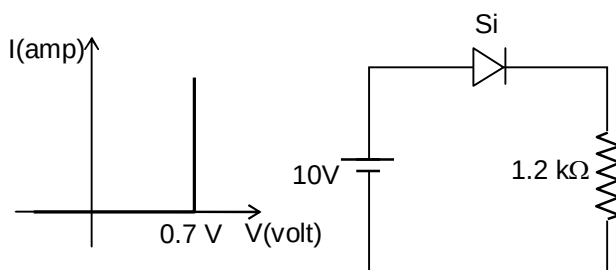
(B) $\frac{32\sqrt{2}}{\pi} \text{ m}$

(C) $\frac{16\sqrt{2}}{\pi} \text{ m}$

(D) $\frac{16}{\pi} \text{ m}$

9. In the figure (a) I-V characteristic of Si-diode is shown. In figure(b) shows a circuit having the Si-diode. The current in the circuit is

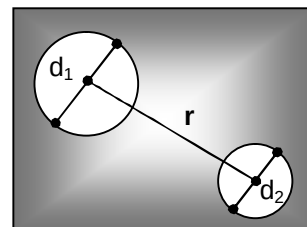
- (A) 8.33 mA
(B) 0
(C) 8.92 mA
(D) 7.75 mA



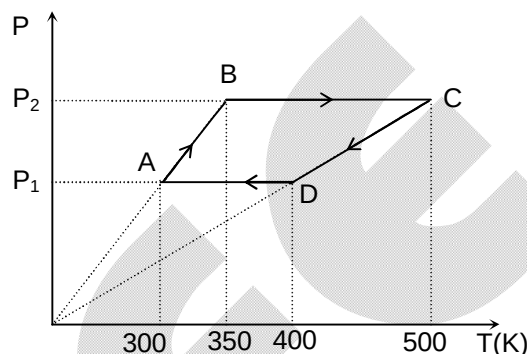
figure(a)

figure(b)

10. Two holes of unequal diameters d_1 and d_2 ($d_2 < d_1$) cut in metal sheet. Separation between the centres of the holes is r . Now the sheet is heated to a higher temperature. Then
- (A) d_1, d_2 increases and r decreases
 (B) d_1, d_2 increases and r increases
 (C) d_1, d_2 decreases and r increases
 (D) d_1, d_2 decreases and r decreases



11. P-T curve of a cyclic process is shown. If number of moles of the monoatomic ideal gas are n . Then efficiency of the cycle is
- (A) 1
 (B) $1/3$
 (C) $1/9$
 (D) $1/11$

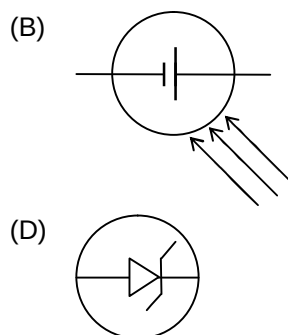
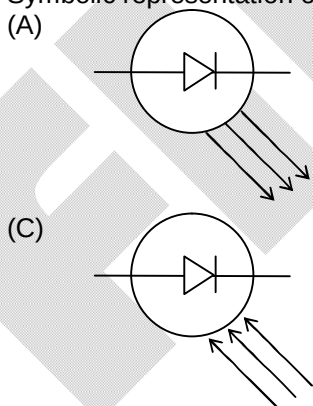


12. In an xyz space only electric field is present having potential function $V = k(x^2 + y^2 + z^2)$ volt, where k is a positive constant and x, y, z are co-ordinate of a point in meter unit. A positive charge particle of charge q and mass m is placed at $(1m, 1m, 1m)$. Acceleration of the particle at the point is

- (A) $\frac{6kq}{m}$
 (B) $\frac{\sqrt{3}kq}{m}$
 (C) $\frac{\sqrt{3}kq}{2m}$
 (D) $\frac{2\sqrt{3}kq}{m}$

13. A nucleus X of mass number 218 in free state decays to emit an α -particle. Kinetic energy of the α -particle emitted is 6.7 MeV. The Q-value of the α -decay nuclear reaction is (in MeV)
- (A) 0.125
 (B) 6.825
 (C) 6.8
 (D) 6.7

14. Symbolic representation of photodiode is

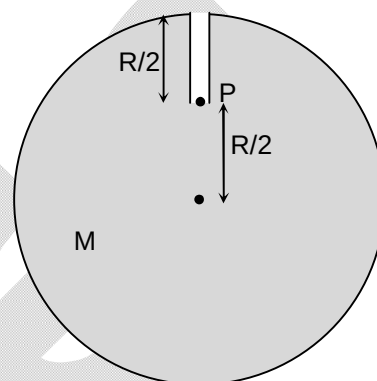


15. Two coherent sound sources in phase of frequencies of 256 Hz and 260 Hz are sounding together. If an observer records an intensity of I_0 at $t = 0$, then he will observe intensity at $t = \frac{1}{16}$ sec.

(A) I_0 (B) $\frac{I_0}{2}$
(C) $\frac{I_0}{4}$ (D) zero

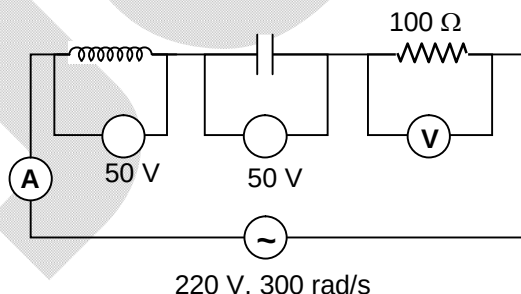
16. A particle is projected from a depth $\frac{R}{2}$ inside the earth such that it escapes to infinity. Mass of the particle is negligibly small is compared to mass of earth M . Then escape speed from the point P is

(A) $\sqrt{\frac{2GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$
(C) $\sqrt{\frac{13GM}{4R}}$ (D) $\sqrt{\frac{11GM}{4R}}$



17. For the AC circuit in the figure, choose the correct option having reading of AC voltmeter (V) and AC ammeter (A).

(A) 120 V, 1.2 A
(B) 220 V, 2.2 A
(C) 320 V, 3.2 A
(D) 220V, 1.2 A



18. The color code of a resistance is given in the figure. The value of resistance and tolerance, respectively are

(A) 47 kΩ, 10%
(B) 4.7 kΩ, 5%
(C) 470 Ω, 5%
(D) 470 kΩ, 5%

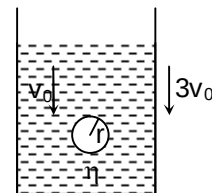


19. A non conducting ring of mass m and radius R a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that the plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0 t^2$ tesla is switched on at $t = 0$. After 2 second for switching on the magnetic field the ring is just about to rotate about vertical axis through its centre. m , R , Q , B are in SI unit. Then

(A) The induced electric field is steady with respect to time in the space
(B) the net electric force acting on the ring at any time t is $QB_R t$
(C) Friction first acts at $t = 2$ sec

(D) The coefficient of static friction between the ring and the surface is $\frac{2B_0 R Q}{mg}$

20. A container filled with viscous liquid is moving vertically downward with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertical downwards in liquid has a speed v_0 with respect to ground. The coefficient of viscosity is η . There is no relative motion between the container and the liquid. Then at the shown instant, the power developed by viscous force with respect to ground frame is



- (A) $-6\pi\eta rv_0^2$ (B) zero
(C) $12\pi\eta rv_0^2$ (D) $-12\pi\eta rv_0^2$

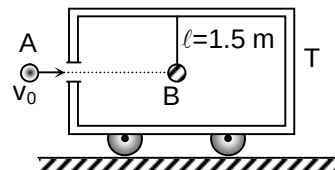
SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

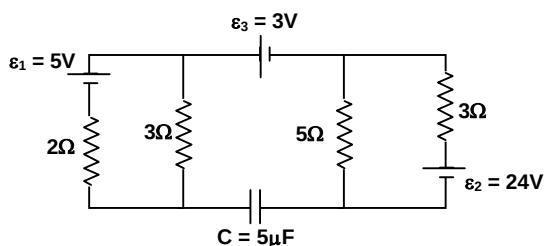
21. An unpolarized light of intensity 80 W/m^2 passes through three polarizing sheets whose pass axes make an angle of 30° , 60° and 30° with y-axis in same sense. The final transmitted intensity after passing through all three polarizers is $I \text{ W/m}^2$. Then find the value of $2I$.
22. A particle of specific charge $2 \times 10^{-2} \text{ C/kg}$ starts moving from the origin under the action of a uniform electric field $\vec{E} = (100 \text{ N/C})\hat{i}$ and magnetic field $\vec{B} = (5\text{T})\hat{k}$. Its velocity vector at $(a, b, 0) \text{ m}$ is $(4\hat{i} - 3\hat{j}) \text{ m/s}$. Then value of $4a$ is (in m)
23. An object is projected with a velocity of 10 m/s making an angle of 30° with horizontal from origin of x-y vertical plane having positive x-axis along horizontal and positive y-axis along the vertically upward direction. The equation of the trajectory is $y = Ax - Bx^2$ where x and y are co-ordinates at a point in meter. Find the value of $\sqrt{3} \left(\frac{A}{B} \right)$ (in SI unit). (Take $g = 10 \text{ m/s}^2$)
24. Each side of a cube is measured to be 7.203 m . If the total surface area of the cube considering the correct significant digits in m^2 is A , then find the value of $10A$.
25. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in 40 seconds is $\tau \text{ N-m}$. Find the value of $\pi\tau$. (Take $\pi^2 = 10$)

26. In the figure shown a ball A of mass $m = 1 \text{ kg}$ collide with another identical ball B with a speed $v_0 = 6 \text{ m/s}$ and sticks to it. The ball B is hanging inside a trolley T of mass $M = 4 \text{ kg}$. Find the maximum height (in cm) attained by the ball (A + B) from its initial level after the collision (Take $g = 10 \text{ m/s}^2$)



27. The solar constant (intensity of solar radiation) just at Venus surface measured to be 2600 W/m^2 . If the radius of the Sun is $7 \times 10^8 \text{ m}$, while the Venus-Sun distance is $1.1 \times 10^{11} \text{ m}$, then find the most probable wavelength in the solar radiation in nm. [Take $(1.81)^{\frac{1}{4}} = 1.16$, wein's constant $b = 2.9 \times 10^{-3} \text{ m-K}$] Sun can be assumed as a black body.

28. If energy stored in the capacitor C in the steady state is $U_0 \mu\text{J}$, then find $2U_0$.



29. A parallel plate capacitor with $A = 1.2 \text{ m}^2$ and separation $d = 1 \text{ mm}$ is charged by a current $i = (8A)e^{-\frac{t}{4\text{sec}}}$. Consider a plane surface X of area 0.6 m^2 parallel to the plates and drawn symmetrically between the plates. Find the displacement current (in mA) through the plane surface X at $t = 4 \text{ sec}$. (Take $e^{-1} = 0.37$)
30. A submarine A travelling at 18 km/hr is being chased along the line of its velocity by another submarine B travelling at 27 km/hr . B send a Sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν (in Hz) is close to (take speed of sound in water = 1500 m/s)

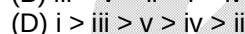
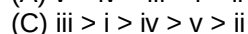
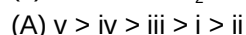
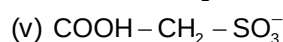
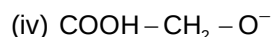
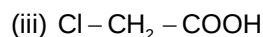
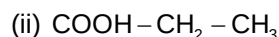
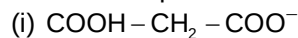
Chemistry

PART – B

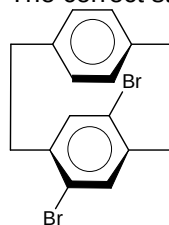
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Compare the acidic strength of following compounds and arrange in decreasing order. Choose the correct option



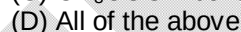
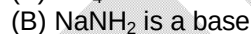
32. The correct statement about the following molecule is:



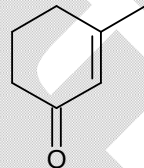
- (A) Molecule is chiral and does not possess chiral centre.
 (B) Molecule is chiral and possess chiral centre.
 (C) Molecule is achiral due to plane of symmetry.
 (D) Molecule is achiral due to axis of symmetry.

33. Choose the correct statement:

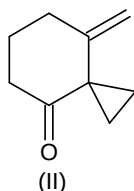
In liquid ammonia:



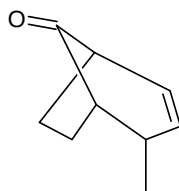
34. The product(s) are :



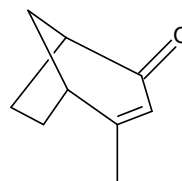
(I)



(II)



(III)



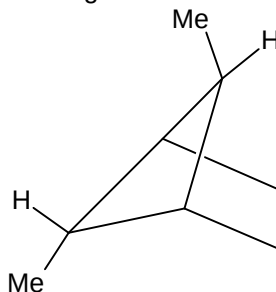
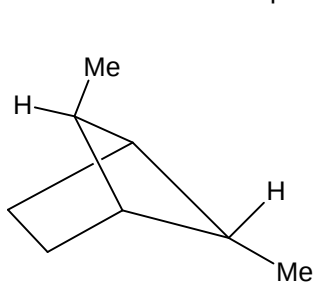
(IV)

- (A) I and III
 (C) I and II

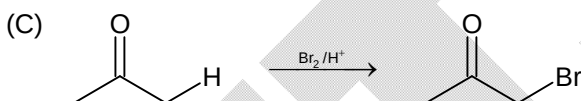
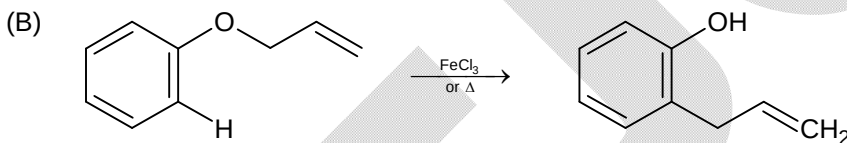
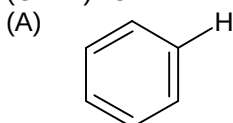
- (B) II and IV
 (D) II and III

35. Considering N as central atom, the structure of $(\text{CH}_3)_2\text{NCS}$ and $(\text{SiH}_3)_2\text{NCS}$ respectively, are:
 (A) Linear and Bent (B) Bent and Bent
 (C) Linear and Linear (D) Bent and linear

36. The correct relationship between the following structures is that they are:



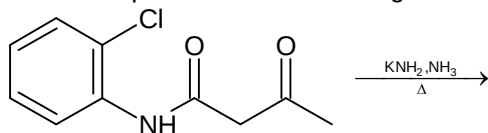
- (A) Enantiomers (B) Identical
 (C) Diastereomers (D) Constitutional isomers
37. The reaction that is expected to show primary kinetic isotope effect for the indicated H-atom (C – H) is:



38. The pair in which both actinides show +3 oxidation state only is:
 (A) Ac and Lr (B) Ac and No
 (C) Cm and Bk (D) Cm and Lr
39. I_2 is violet in the solid as well as in gas phase. However, in acetone or ethanol, it turns brown. Choose the correct statement(s) for this colour change:
 (i) Dissociation of I_2 in atomic state.
 (ii) Interaction of low-lying σ^* orbital of iodine with lone pair of O (solvent).
 (iii) Formation of a charge transfer complex.
 (A) (i) only (B) (ii) only
 (C) (i) and (ii) only (D) (iii) only

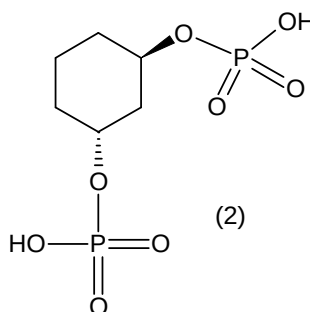
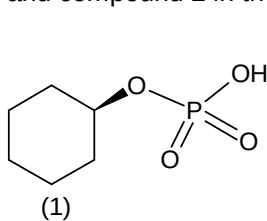
40. An atomic orbital is given by $nr^2 e^{-\frac{r}{3a_0}} (3\cos^2\theta - 1)$ in hydrogen atom. The orbital represents:
 (A) 2p (B) 3p
 (C) 3d (D) 4d

41. The magnitude of bond angles in gaseous NF_3 , SbF_3 and SbCl_3 follows the order:
 (A) $\text{NF}_3 > \text{NH}_3 > \text{SbCl}_3$ (B) $\text{SbCl}_3 > \text{NH}_3 > \text{NF}_3$
 (C) $\text{NH}_3 > \text{NF}_3 > \text{SbCl}_3$ (D) $\text{NF}_3 > \text{SbCl}_3 > \text{NH}_3$
42. Consider the following chemical reactions in gas phase
 Reaction I : $\text{H} + \text{D}_2 \longrightarrow \text{HD} + \text{D}$
 Reaction II : $\text{D} + \text{H}_2 \longrightarrow \text{HD} + \text{H}$
 Which of the following statement is true?
 (A) Reaction I and II, both are endothermic.
 (B) Reaction I is endothermic, Reaction II is exothermic.
 (C) Both reactions are thermo neutral as they involve isotopes which differ by the number of neutrons, which do not play any role in chemical reactions.
 (D) Reaction I and II both are exothermic.
43. Predict the product of the following reaction

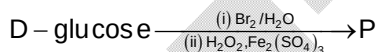


- (A)
- (B)
- (C)
- (D)

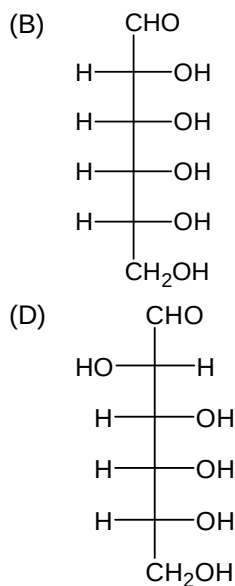
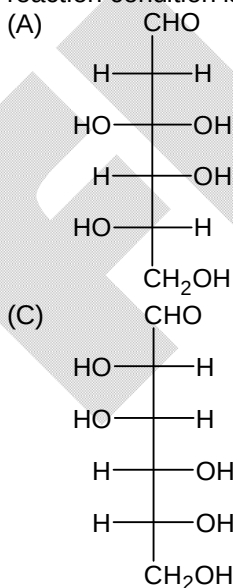
44. Compound 2 and compound 1 were mixed in a 1 : 4 molar ratio. The mixture was subjected to an ammonium molybdate test for phosphate estimation. The result of the molybdated test yielded a phosphate concentration of 36.0 mM for the mixture. What are the concentration of compound 1 and compound 2 in the mixture?



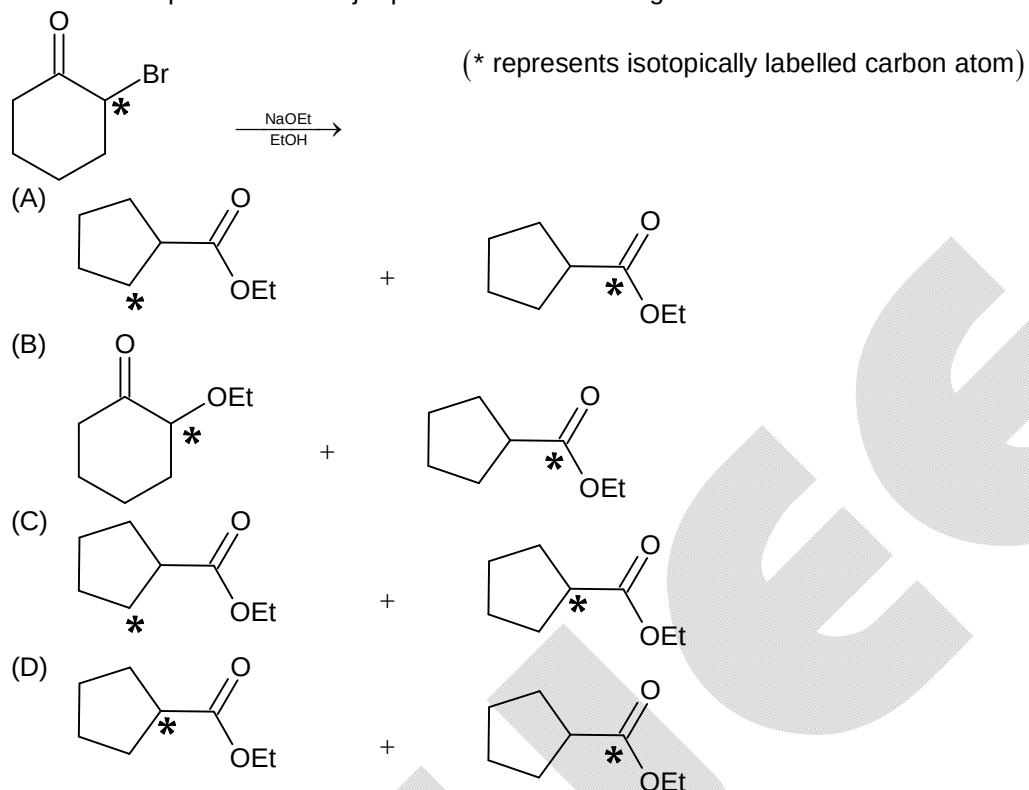
- (A) Compound 1 : 30 mM Compound 2 : 6 mM
 (B) Compound 1 : 24 mM Compound 2 : 6 mM
 (C) Compound 1 : 28.8 mM Compound 2 : 7.2 mM
 (D) Compound 1 : 24 mM Compound 2 : 12 mM
45. Which of the following can't be used for the given transformation?
- $$\text{H}_3\text{C}-\text{C}(=\text{O})-\text{C}(\text{CH}_3)_3 \longrightarrow \text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)_3$$
- (A) Clemmenson reduction (B) Wolf-Kishner reduction
 (C) Mozingo reduction (D) Bouveault-Blanc reduction
46. A mixture of acetone and CCl_4 can be separated by
- (A) Azeotropic distillation (B) Fractional distillation
 (C) Steam distillation (D) Vacuum distillation
47. In the following reaction of D-(+)glucose a product P is formed.



Amongst the following compounds, the one which will give the same product (P) under identical reaction condition is:



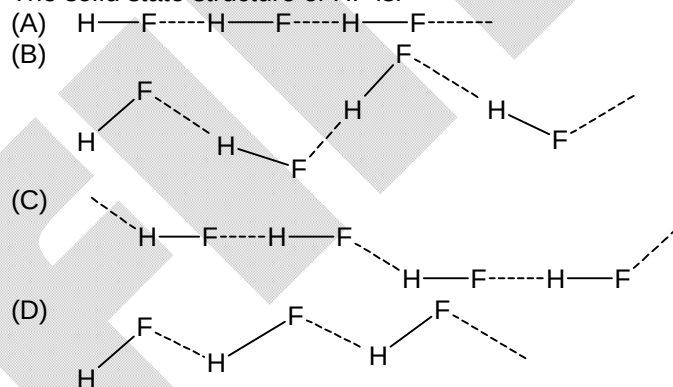
48. The correct option for the major product of the following reaction is:



49. The salt bridge in a Galvanic cell allows the flow of:

(A) ions but not electrons
 (B) both ions and electrons
 (C) electrons but not ions
 (D) neither ions nor electrons

50. The solid state structure of HF is:

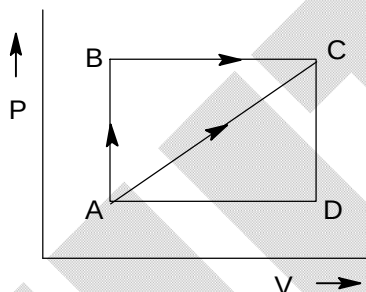


SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The number of stereoisomers for the complex $[\text{RhH}(\text{C} \equiv \text{CR})_2(\text{PMe}_3)_3]$ is:
52. The total number of lone pairs of electron on all atoms in cyanogens is 'x' and thiocyanogen is 'y'. Report $(x + y) = ?$
53. For the cell $\text{Cd} | \text{CdCl}_2 || \text{AgCl} | \text{Ag}$; $E^\circ_{\text{cell}} = 0.675 \text{ V}$ and $\left(\frac{dE^\circ}{dT}\right) = -6.5 \times 10^{-4} \text{ VK}^{-1}$ at 27°C . The $\Delta H^\circ (\text{kJ mol}^{-1})$ value for the cell is:
54. An organic compound is estimated through Dumas method and was found to evolve 264 g CO_2 , 72 g of H_2O and 28 g of N_2 . If the formula of the compound is $\text{C}_x\text{H}_y\text{N}_z$. Then $\left(\frac{x+y}{z}\right)$ is:
55. When glycinium hydrochloride is titrated against NaOH, pH at the first half equivalence point is 2.40 and the pH at the second half equivalence point is 9.60. The pH at first equivalence point is:
56. A thermodynamic process is shown below:
 Given: $P_A = 3 \times 10^4 \text{ Pa}$, $P_B = 8 \times 10^4 \text{ Pa}$, $V_A = 2 \times 10^{-3} \text{ m}^3$, $V_D = 5 \times 10^{-3} \text{ m}^3$. In the process AB, 600 J of heat is added and in BC 200 J of heat is added to the system. If the change in internal energy of the system in the process AC is x; then report $\left(\frac{x}{20}\right)$. x is calculated in Joule.



57. 10.6 g of benzaldehyde was reacted with an excess of acetophenone to produce 10.4 g of the enone product according to the given equation. Calculate the % yield of the product.

$$\text{Benzaldehyde} + \text{Acetophenone} \xrightarrow[2. \Delta]{1. \text{NaOEt}} \text{Product}$$
58. Assume that the reaction of MeMgBr with ethyl acetate proceeds with 100% conversion to tert – butanol. The volume of 0.2 M solution of MeMgBr required to convert 10 mL of a 0.025 M solution of ethyl acetate is x. Report the value of $10x$ in mL.

59. The mechanism of enzyme catalysis is as follows:



The rate of product formation is given as:

$$\frac{+dP}{dt} = \frac{k_1 k_2 [E_0][S]}{k_{-1} + k_2 + k_1 [S]}, \text{ where } [E_0] \text{ is the total enzyme concentration, } [S] \text{ is substrate concentration.}$$

The rate of product formation when the substrate concentration is very large has a limiting value of 0.02 Ms^{-1} . At a substrate conc. of 250 mg/dm^3 , the rate is half this value. The value of $\frac{k_1}{k_{-1}}$ (in dm^3 / kg) assuming that $k_2 \ll k_{-1}$ is $x \times 10^3$. What is the value of x ?

60. The maximum concentration of equimolar solution of ferrous sulphate and potassium sulphide so that when mixed in equal volumes, there is no precipitate of iron sulphide is $C \times 10^{-9} \text{ M}$. Report the value of C . [Given: $K_{sp}(\text{FeS}) = 6.3 \times 10^{-18} \text{ M}^2$.

Mathematics**PART – C****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If a non constant function $f(x) = \frac{2x^2 - kx + 5}{x^2 + 3x + 4}$ is bounded, then k can not have values is equal to
 (A) -2 (B) -6
 (C) 2 (D) 6
62. If $\sqrt{2} < \alpha, \beta < \sqrt{3}$, $\alpha \neq \beta$, then the value of $\left[\frac{\alpha - \beta}{\cot^{-1} \alpha - \cot^{-1} \beta} \right]$ is (where [.] denotes G.I.F.)
 (A) -4 (B) 4
 (C) 3 (D) -5
63. Let real number a satisfy $\{a\} + \left\{\frac{1}{a}\right\} = 1$ and b is the value of $\{a^3\} + \left\{\frac{1}{a^3}\right\}$ (where {.] denotes fractional part function), then the value of $\int_0^b \frac{x}{1+x} \sqrt{1-x^2} dx$ is
 (A) $2 + \frac{\pi}{4}$ (B) $2 - \frac{\pi}{4}$
 (C) $1 - \frac{\pi}{4}$ (D) $1 + \frac{\pi}{4}$
64. For a real number 'a' such that the equation $a(\sin x + \sin(2x)) = \sin 3x$ has more than one solution in the interval $(0, \pi)$ the set of all such value of a is
 (A) $(-3, -2)$ (B) $(-3, 1) - \{-2\}$
 (C) $(-2, 1)$ (D) $(0, 1)$
65. Let c be a real number and let z_1 and z_2 be the two complex number satisfying the equation $z^2 - cz + 10 = 0$, points $z_1, z_2, \frac{1}{z_1}$ and $\frac{1}{z_2}$ are vertices of (convex) quadrilateral in the complex plane, when the area is maximum then the value of c is
 (A) $4\sqrt{5}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{5}$ (D) $8\sqrt{5}$
66. If the equation $\frac{\alpha^2}{1 - \tan^2 x} = \frac{\sin^2 x + \alpha^2 - 2}{\cos 2x}$ has a solution, then
 (A) $\alpha^2 < 1$ (B) $\alpha^2 > 2$
 (C) $\alpha^2 \geq 1$ (D) $\alpha \in \mathbb{R}$
67. A polynomial of degree four with leading coefficient 1 and integer coefficients, has two zeros, both of which are integer. Which of the following can also be a zero of the polynomial ?
 (A) $\frac{1+i\sqrt{11}}{2}$ (B) $\frac{1+i}{2}$
 (C) $\frac{1}{2} + i$ (D) $1 + \frac{i}{2}$

68. Circle $x^2 + y^2 - 6x - 4y + 11 = 0$ is rolled up by 4 units along a tangent to it at the point (2, 3) let this be circle C_1 , C_2 is the mirror image of circle C_1 about the tangent. A and B are centres of circles C_1 and C_2 , C and D are the feet of perpendicular from A and B respectively upto x-axis, then the area of trapezium ABCD equals to
- (A) $(2 + 2\sqrt{2})$ (B) $(3 + 2\sqrt{2})$
 (C) $(6 + 4\sqrt{2})$ (D) $(4 + 4\sqrt{2})$
69. A point P is selected at random from the interior of the pentagon with vertices $A = (0, 2)$, $B = (4, 0)$, $C = (2\pi + 1, 0)$, $D = (2\pi + 1, 4)$ and $E = (0, 4)$. What is the probability that $\angle APB$ is obtuse ?
- (A) $1/5$ (B) $1/4$
 (C) $3/8$ (D) $5/16$
70. How many 15 letter arrangements of 5 A's, 5B's and 5 C's have no A's in the first 5 letter no B's in the next 5 letters and no C's in the last 5 letters
- (A) $\sum_{k=0}^5 ({}^5C_k)^3$ (B) $\sum_{k=0}^5 ({}^5P_k)^3$
 (C) 2^{15} (D) $2^5 \cdot 3^5$
71. The value of $\sum_{i=0}^n \sum_{j=1}^n {}^nC_j \cdot {}^jC_i$ $i \leq j$
- (A) 3^n (B) $3^n + 1$
 (C) $3^n - 1$ (D) 3^{n+1}
72. A parabola $y = ax^2 + bx + c$ is reflected about the x-axis, the parabola and its reflection are translated horizontally five units in opposite direction to become the graph of $y = f(x)$ and $y = g(x)$ respectively. Then which of the following describes the graph of $y = f(x) + g(x)$
- (A) A horizontal line (B) the graph of a cubic function
 (C) a non horizontal line (D) A parabola
73. If z_1 and z_2 are two complex number such that $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ where $z_1 = 4 + 6i$ and $z_2 = 3 + 5i$, then locus of z is
- (A) straight line with slope $-\frac{1}{2}$ (B) circle with radius $\frac{1}{\sqrt{2}}$
 (C) hyperbola with eccentricity $\sqrt{2}$ (D) hyperbola with eccentricity $\frac{5}{2}$
74. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$, then $\int_{1/10}^{9/10} f(f(x))dx$ is equal to
- (A) $\frac{4}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{5}$ (D) 0
75. If $I_1 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}} x^2 dx$ and $I_2 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}+1} x^2 dx$, then $\frac{I_1 - I_2}{I_2} = \frac{1}{k}$, then k is equal to
- (A) $(\sqrt{3} + 1)\sqrt{2}$ (B) $(\sqrt{3} - 1)\sqrt{2}$
 (C) $(\sqrt{3} + \sqrt{2})$ (D) $3(\sqrt{6} + \sqrt{2})$

76. Total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^{10} + 1}{t^{10} + t^4 + 1} dt = 6x - 5$ is
 (A) 0 (B) 1
 (C) 2 (D) 3
77. A continuous and differentiable function $f(x)$ satisfy the functional equation $f(x + y) = \frac{f(x)f(y)}{f(x) + f(y)}$,
 $f(1) = 32$ then $f(4) =$
 (A) 8 (B) 6
 (C) 4 (D) 2
78. $\lim_{x \rightarrow \infty} |1 - a^4|^x \sin \frac{5}{|1 - a^2|^x} a^2 > 2$ is
 (A) 0 (B) 5
 (C) ∞ (D) does not exist
79. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2 - x)$ and $f''(x) < 0 \forall x \in (0, 2)$ then which is true ?
 (A) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $(0, 2)$
 (B) $g(x)$ increases in $(0, 2)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (C) $g(x)$ increases in $\left(\frac{4}{3}, 2\right)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (D) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $\left(\frac{4}{3}, 2\right)$
80. Area bounded by the curve $y = 1$ and $y = \frac{\sin x + \cos x + |\sin x - \cos x|}{2}$ for $x \in [0, 2\pi]$ is
 (A) $2\pi - \sqrt{2}$ (B) $2\pi - 2\sqrt{2}$
 (C) $2(\pi - \sqrt{2} - 1)$ (D) $2(\pi + \sqrt{2} - 3)$

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = \sqrt{ax^2 + bx}$, for how many real values of a is there at least one positive value of b for which the domain of $f(x)$ and the range of $f(x)$ are the same set
82. Lines $x + y + 4 = 0$, $2x + ky + 2 = 0$ intersects the circle $x^2 + y^2 + 2x + 4y + 1 = 0$ and $x^2 + y^2 + 4x + 6y + 4 = 0$ as 4 points which are concyclic, then the value of k is
83. The number of distinct values of determinant of order 2×2 whose entries are from the set $\{-1, 0, 1\}$ is
84. Krishna writes the number 1, 2, 3, ..., 9 on separate cards, one number per card, he wishes to divide the cards into 3 groups of 3 cards so that the sum of the numbers in each group will be the same. In how many ways can this be done ?

85. Fifteen integers $a_1, a_2, a_3, \dots, a_{15}$ are arranged in order on a number line. The integers are equally spaced and have the property that $1 \leq a_1 \leq 10$, $13 \leq a_2 \leq 20$ and $241 \leq a_{15} \leq 250$. What is the sum of digits of a_{14} ?
86. In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term is 4000. What is the first term
87. A data set consists of 6 (not all distinct) positive integers : 1, 7, 5, 2, 5 and x, the Arithmetic Mean of the 6 numbers equals a value in the data set. What is the sum of all possible values of x
88. The area bounded by the graph of $y^2 + 2xy + 40|x| = 400$ is
89. Let the distance between two point (x_1, y_1) and (x_2, y_2) is given by $|x_1 - x_2| + |y_1 - y_2|$ for how many points P with integer coordinates is the distance between P and the origin less than or equal to 20
90. Two integers are inserted into list 3, 3, 3, 8, 11, 12, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of the two additional numbers

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2024

TEST DATE: 14-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

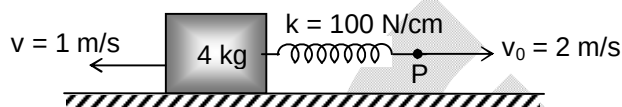
Physics

PART – A

SECTION – A (One Options Correct Type)

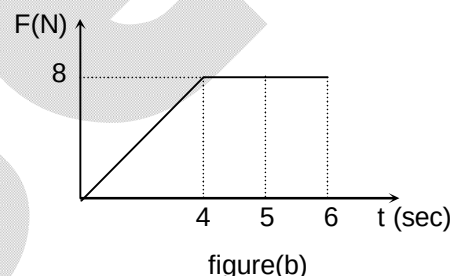
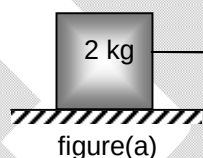
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A spring block system lies on a smooth horizontal surface. The free end of the spring is being pulled towards right with constant speed $v_0 = 2 \text{ m/s}$. At $t = 0$, the spring (spring constant $k = 100 \text{ N/cm}$) is unstretched and block is having a speed $v = 1 \text{ m/s}$ to the left. The maximum acceleration of the block during the motion is
- (A) 50 m/s^2
(C) 150 m/s^2



- (B) 100 m/s^2
(D) 200 m/s^2

2. A block of mass 2 kg is kept on a rough horizontal surface having coefficient of friction ($\mu_s = 0.3$ and $\mu_k = 0.25$). A time dependent force F is applied on the block as shown in the figure (b). Find the speed of the block at $t = 5 \text{ sec}$.



- (A) 12 m/s
(C) 2.5 m/s

- (B) zero
(D) 0.5 m/s

3. A container of fixed volume has mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T . Assuming the gases are ideal, then choose the INCORRECT Statement.
- (A) Internal energy of the mixture is $4RT$ Joule

- (B) Ratio of rms speed of gas molecules in the mixture and hydrogen is $\sqrt{\frac{2}{3}}$

- (C) Ratio of speed of sound in the mixture and helium is $\sqrt{\frac{6}{5}}$

- (D) Ratio of rms speed of the gas molecules in Helium and hydrogen is $\sqrt{2}$

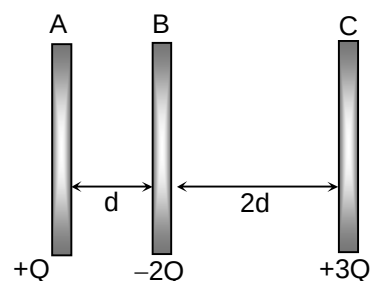
4. Three large metallic parallel plates A, B and C each having area A are separated by very small distances as shown in the figure. Charges Q , $-2Q$ and $3Q$ are given to the plates. The potential difference between plates C and A is

- (A) zero

- (B) $\frac{2Qd}{\epsilon_0 A}$

- (C) $\frac{4Qd}{\epsilon_0 A}$

- (D) Indeterminate

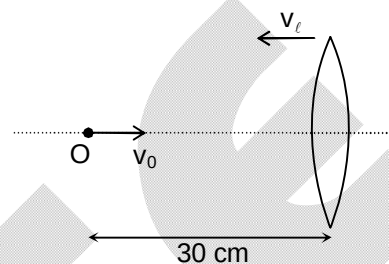


5. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$ and comes out without any change in magnitude or direction of $\vec{v}$. Then (here $\vec{E}$ is electric field and $\vec{B}$ is magnetic field)

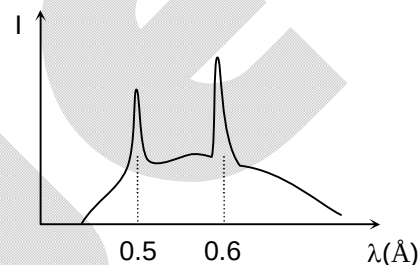
(A) $\vec{B} \times \vec{E} = |\vec{E}|^2 \vec{v}$ (B) $\vec{E} \times \vec{B} = |\vec{B}|^2 \vec{v}$
 (C) $\vec{B} \times \vec{E} = |\vec{B}|^2 \vec{v}$ (D) $\vec{E} \times \vec{B} = |\vec{E}|^2 \vec{v}$

6. A point object has velocity $v_0 = 2$ cm/s when it is placed at a distance of 30 cm from a lens moving with a speed $v_\ell = 2$ cm/s towards left. If the focal length of the lens is $f = 20$ cm, find the velocity of the image with respect to the ground.

- (A) 14 cm/s towards right
 (B) 14 cm/s towards left
 (C) 25 cm/s towards right
 (D) 25 cm/s towards left



7. The graph shows the intensity of X-rays emitted by a heavy target and the wavelength, when electrons of energy 20 keV are incident on the target. In the graph one peak is of K_α line and the other peak is of K_β line. Then



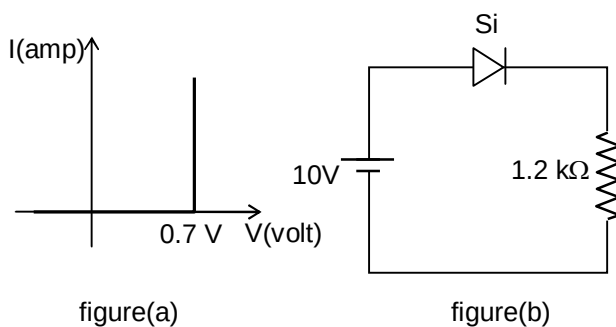
- (A) If the energy of incident electron is increased to 30 keV, then both the peaks will shift rightward.
 (B) If the energy of incident electron is increased to 30 keV, then both the peaks will shift leftward.
 (C) K_α line wavelength is 0.5 Å
 (D) K_α line wavelength is 0.6 Å

8. A 2 kg particle is moving along the x-axis under the action of the force $\vec{F} = -\frac{\pi^2}{32} \vec{x}$, where F is in newton and x is in meter. At $t = 2$ sec, the particle is moving along $\vec{x}$ and is at $x = 0$. At $t = 10$ sec it is observed that particle's speed is $4\sqrt{2}$ m/s. Then amplitude of particle motion is

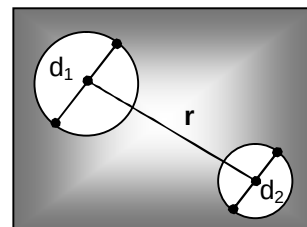
(A) $\frac{32}{\pi}$ m (B) $\frac{32\sqrt{2}}{\pi}$ m
 (C) $\frac{16\sqrt{2}}{\pi}$ m (D) $\frac{16}{\pi}$ m

9. In the figure (a) I-V characteristic of Si-diode is shown. In figure(b) shows a circuit having the Si-diode. The current in the circuit is

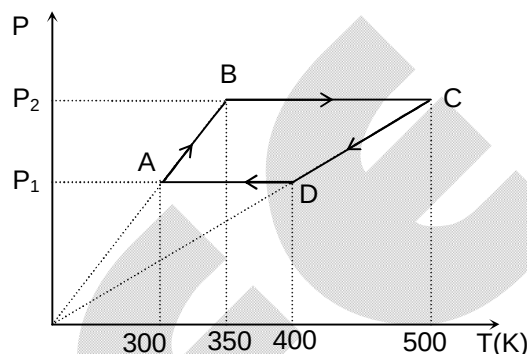
- (A) 8.33 mA
 (B) 0
 (C) 8.92 mA
 (D) 7.75 mA



10. Two holes of unequal diameters d_1 and d_2 ($d_2 < d_1$) cut in metal sheet. Separation between the centres of the holes is r . Now the sheet is heated to a higher temperature. Then
- (A) d_1, d_2 increases and r decreases
 (B) d_1, d_2 increases and r increases
 (C) d_1, d_2 decreases and r increases
 (D) d_1, d_2 decreases and r decreases



11. P-T curve of a cyclic process is shown. If number of moles of the monoatomic ideal gas are n . Then efficiency of the cycle is
- (A) 1
 (B) $1/3$
 (C) $1/9$
 (D) $1/11$

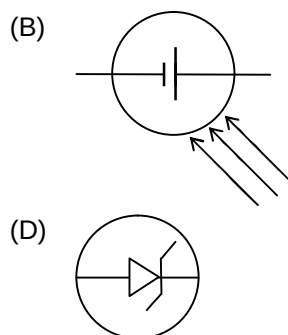
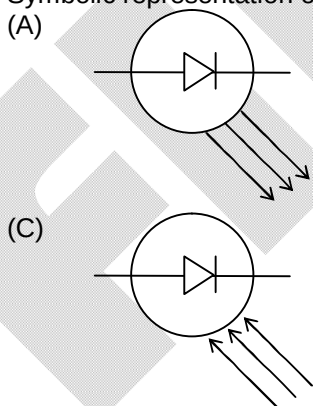


12. In an xyz space only electric field is present having potential function $V = k(x^2 + y^2 + z^2)$ volt, where k is a positive constant and x, y, z are co-ordinate of a point in meter unit. A positive charge particle of charge q and mass m is placed at $(1m, 1m, 1m)$. Acceleration of the particle at the point is

- (A) $\frac{6kq}{m}$
 (B) $\frac{\sqrt{3}kq}{m}$
 (C) $\frac{\sqrt{3}kq}{2m}$
 (D) $\frac{2\sqrt{3}kq}{m}$

13. A nucleus X of mass number 218 in free state decays to emit an α -particle. Kinetic energy of the α -particle emitted is 6.7 MeV. The Q-value of the α -decay nuclear reaction is (in MeV)
- (A) 0.125
 (B) 6.825
 (C) 6.8
 (D) 6.7

14. Symbolic representation of photodiode is

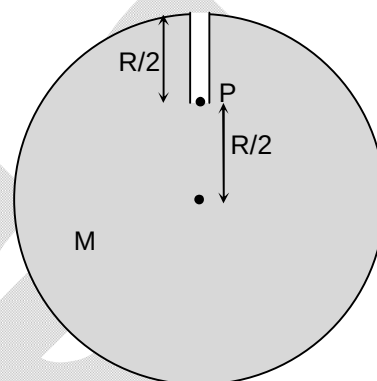


15. Two coherent sound sources in phase of frequencies of 256 Hz and 260 Hz are sounding together. If an observer records an intensity of I_0 at $t = 0$, then he will observe intensity at $t = \frac{1}{16}$ sec.

(A) I_0 (B) $\frac{I_0}{2}$
(C) $\frac{I_0}{4}$ (D) zero

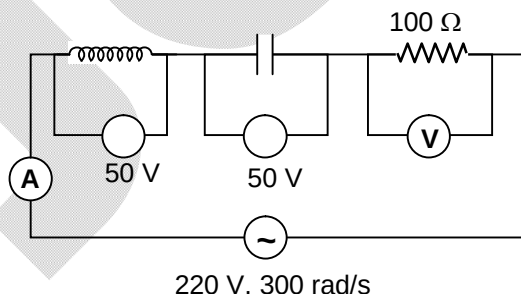
16. A particle is projected from a depth $\frac{R}{2}$ inside the earth such that it escapes to infinity. Mass of the particle is negligibly small is compared to mass of earth M . Then escape speed from the point P is

(A) $\sqrt{\frac{2GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$
(C) $\sqrt{\frac{13GM}{4R}}$ (D) $\sqrt{\frac{11GM}{4R}}$



17. For the AC circuit in the figure, choose the correct option having reading of AC voltmeter (V) and AC ammeter (A).

(A) 120 V, 1.2 A
(B) 220 V, 2.2 A
(C) 320 V, 3.2 A
(D) 220V, 1.2 A



18. The color code of a resistance is given in the figure. The value of resistance and tolerance, respectively are

(A) 47 kΩ, 10%
(B) 4.7 kΩ, 5%
(C) 470 Ω, 5%
(D) 470 kΩ, 5%

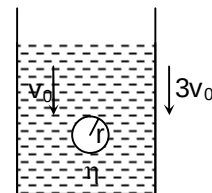


19. A non conducting ring of mass m and radius R a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that the plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0 t^2$ tesla is switched on at $t = 0$. After 2 second for switching on the magnetic field the ring is just about to rotate about vertical axis through its centre. m, R, Q, B are in SI unit. Then

(A) The induced electric field is steady with respect to time in the space
(B) the net electric force acting on the ring at any time t is $QB_R t$
(C) Friction first acts at $t = 2$ sec

(D) The coefficient of static friction between the ring and the surface is $\frac{2B_0 R Q}{mg}$

20. A container filled with viscous liquid is moving vertically downward with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertical downwards in liquid has a speed v_0 with respect to ground. The coefficient of viscosity is η . There is no relative motion between the container and the liquid. Then at the shown instant, the power developed by viscous force with respect to ground frame is



- (A) $-6\pi\eta rv_0^2$ (B) zero
(C) $12\pi\eta rv_0^2$ (D) $-12\pi\eta rv_0^2$

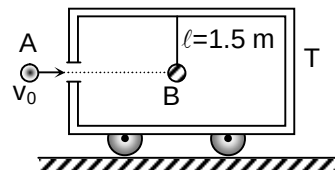
SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

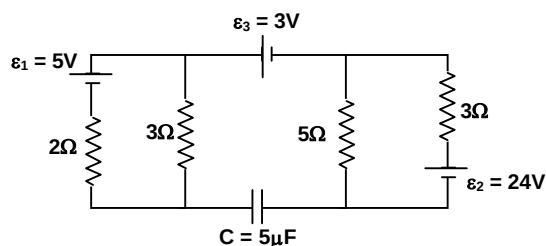
21. An unpolarized light of intensity 80 W/m^2 passes through three polarizing sheets whose pass axes make an angle of 30° , 60° and 30° with y-axis in same sense. The final transmitted intensity after passing through all three polarizers is $I \text{ W/m}^2$. Then find the value of $2I$.
22. A particle of specific charge $2 \times 10^{-2} \text{ C/kg}$ starts moving from the origin under the action of a uniform electric field $\vec{E} = (100 \text{ N/C})\hat{i}$ and magnetic field $\vec{B} = (5\text{T})\hat{k}$. Its velocity vector at $(a, b, 0) \text{ m}$ is $(4\hat{i} - 3\hat{j}) \text{ m/s}$. Then value of $4a$ is (in m)
23. An object is projected with a velocity of 10 m/s making an angle of 30° with horizontal from origin of x-y vertical plane having positive x-axis along horizontal and positive y-axis along the vertically upward direction. The equation of the trajectory is $y = Ax - Bx^2$ where x and y are co-ordinates at a point in meter. Find the value of $\sqrt{3} \left(\frac{A}{B} \right)$ (in SI unit). (Take $g = 10 \text{ m/s}^2$)
24. Each side of a cube is measured to be 7.203 m . If the total surface area of the cube considering the correct significant digits in m^2 is A , then find the value of $10A$.
25. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in 40 seconds is $\tau \text{ N-m}$. Find the value of $\pi\tau$. (Take $\pi^2 = 10$)

26. In the figure shown a ball A of mass $m = 1 \text{ kg}$ collide with another identical ball B with a speed $v_0 = 6 \text{ m/s}$ and sticks to it. The ball B is hanging inside a trolley T of mass $M = 4 \text{ kg}$. Find the maximum height (in cm) attained by the ball (A + B) from its initial level after the collision (Take $g = 10 \text{ m/s}^2$)



27. The solar constant (intensity of solar radiation) just at Venus surface measured to be 2600 W/m^2 . If the radius of the Sun is $7 \times 10^8 \text{ m}$, while the Venus-Sun distance is $1.1 \times 10^{11} \text{ m}$, then find the most probable wavelength in the solar radiation in nm. [Take $(1.81)^{\frac{1}{4}} = 1.16$, wein's constant $b = 2.9 \times 10^{-3} \text{ m-K}$] Sun can be assumed as a black body.

28. If energy stored in the capacitor C in the steady state is $U_0 \mu\text{J}$, then find $2U_0$.



29. A parallel plate capacitor with $A = 1.2 \text{ m}^2$ and separation $d = 1 \text{ mm}$ is charged by a current $i = (8A)e^{-\frac{t}{4\text{sec}}}$. Consider a plane surface X of area 0.6 m^2 parallel to the plates and drawn symmetrically between the plates. Find the displacement current (in mA) through the plane surface X at $t = 4 \text{ sec}$. (Take $e^{-1} = 0.37$)
30. A submarine A travelling at 18 km/hr is being chased along the line of its velocity by another submarine B travelling at 27 km/hr . B send a Sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν (in Hz) is close to (take speed of sound in water = 1500 m/s)

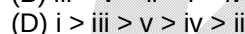
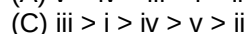
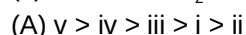
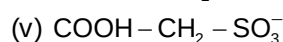
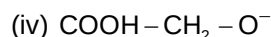
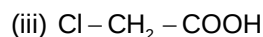
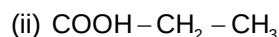
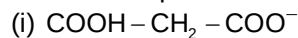
Chemistry

PART – B

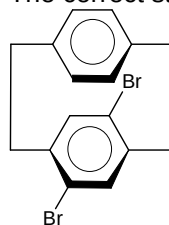
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Compare the acidic strength of following compounds and arrange in decreasing order. Choose the correct option



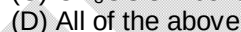
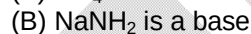
32. The correct statement about the following molecule is:



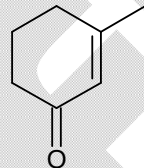
- (A) Molecule is chiral and does not possess chiral centre.
 (B) Molecule is chiral and possess chiral centre.
 (C) Molecule is achiral due to plane of symmetry.
 (D) Molecule is achiral due to axis of symmetry.

33. Choose the correct statement:

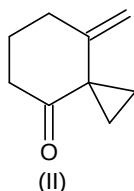
In liquid ammonia:



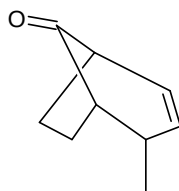
34. The product(s) are :



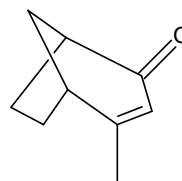
(I)



(II)



(III)



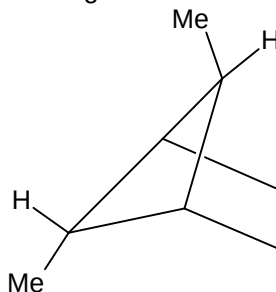
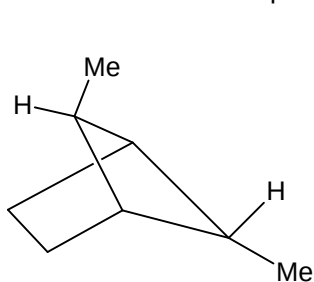
(IV)

- (A) I and III
 (C) I and II

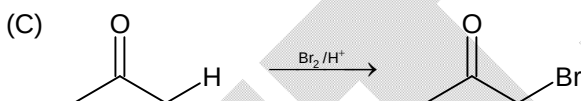
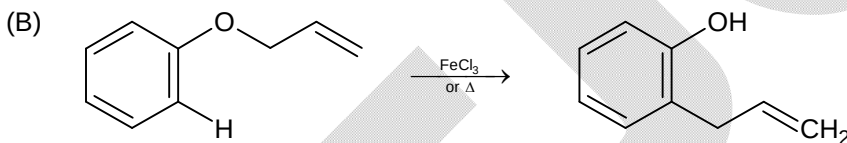
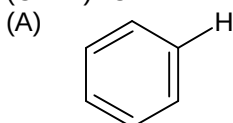
- (B) II and IV
 (D) II and III

35. Considering N as central atom, the structure of $(\text{CH}_3)_2\text{NCS}$ and $(\text{SiH}_3)_2\text{NCS}$ respectively, are:
 (A) Linear and Bent (B) Bent and Bent
 (C) Linear and Linear (D) Bent and linear

36. The correct relationship between the following structures is that they are:



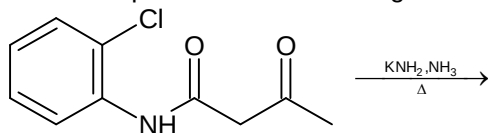
- (A) Enantiomers (B) Identical
 (C) Diastereomers (D) Constitutional isomers
37. The reaction that is expected to show primary kinetic isotope effect for the indicated H-atom (C – H) is:



38. The pair in which both actinides show +3 oxidation state only is:
 (A) Ac and Lr (B) Ac and No
 (C) Cm and Bk (D) Cm and Lr
39. I_2 is violet in the solid as well as in gas phase. However, in acetone or ethanol, it turns brown. Choose the correct statement(s) for this colour change:
 (i) Dissociation of I_2 in atomic state.
 (ii) Interaction of low-lying σ^* orbital of iodine with lone pair of O (solvent).
 (iii) Formation of a charge transfer complex.
 (A) (i) only (B) (ii) only
 (C) (i) and (ii) only (D) (iii) only

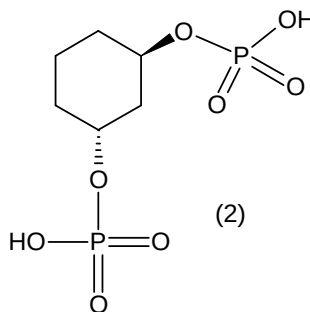
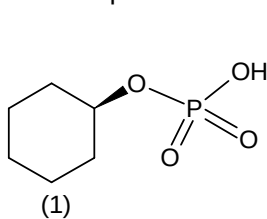
40. An atomic orbital is given by $nr^2 e^{-\frac{r}{3a_0}} (3\cos^2\theta - 1)$ in hydrogen atom. The orbital represents:
 (A) 2p (B) 3p
 (C) 3d (D) 4d

41. The magnitude of bond angles in gaseous NF_3 , SbF_3 and SbCl_3 follows the order:
 (A) $\text{NF}_3 > \text{NH}_3 > \text{SbCl}_3$ (B) $\text{SbCl}_3 > \text{NH}_3 > \text{NF}_3$
 (C) $\text{NH}_3 > \text{NF}_3 > \text{SbCl}_3$ (D) $\text{NF}_3 > \text{SbCl}_3 > \text{NH}_3$
42. Consider the following chemical reactions in gas phase
 Reaction I : $\text{H} + \text{D}_2 \longrightarrow \text{HD} + \text{D}$
 Reaction II : $\text{D} + \text{H}_2 \longrightarrow \text{HD} + \text{H}$
 Which of the following statement is true?
 (A) Reaction I and II, both are endothermic.
 (B) Reaction I is endothermic, Reaction II is exothermic.
 (C) Both reactions are thermo neutral as they involve isotopes which differ by the number of neutrons, which do not play any role in chemical reactions.
 (D) Reaction I and II both are exothermic.
43. Predict the product of the following reaction

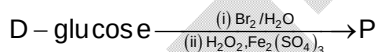


- (A)
- (B)
- (C)
- (D)

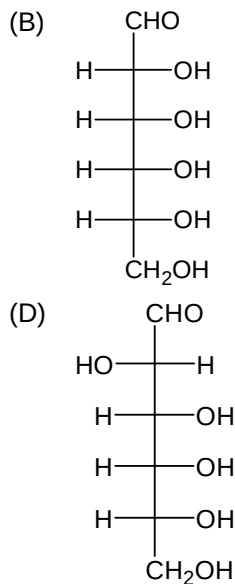
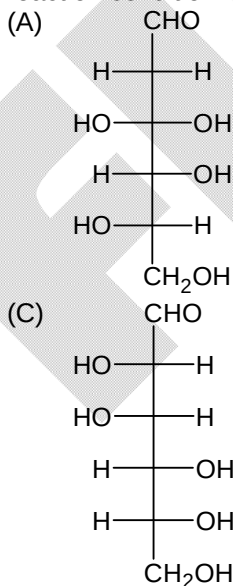
44. Compound 2 and compound 1 were mixed in a 1 : 4 molar ratio. The mixture was subjected to an ammonium molybdate test for phosphate estimation. The result of the molybdated test yielded a phosphate concentration of 36.0 mM for the mixture. What are the concentration of compound 1 and compound 2 in the mixture?



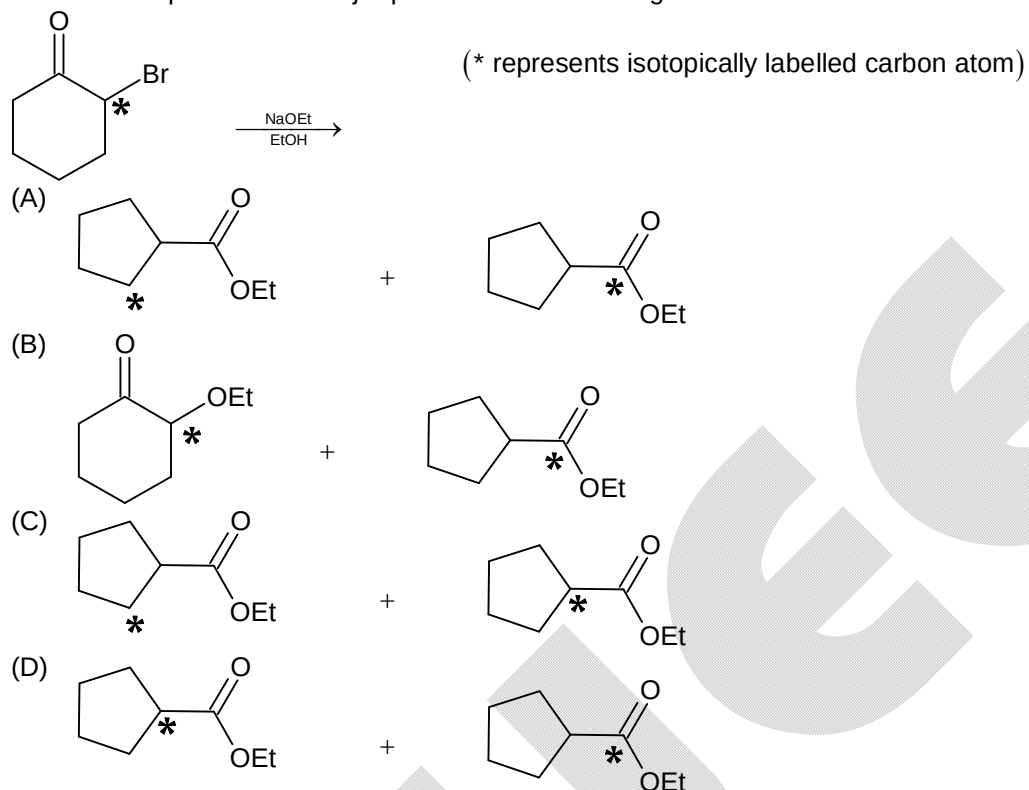
- (A) Compound 1 : 30 mM Compound 2 : 6 mM
 (B) Compound 1 : 24 mM Compound 2 : 6 mM
 (C) Compound 1 : 28.8 mM Compound 2 : 7.2 mM
 (D) Compound 1 : 24 mM Compound 2 : 12 mM
45. Which of the following can't be used for the given transformation?
- $\text{CH}_3 - \text{CH}_2 - \text{C}(\text{CH}_3)_3$
- (A) Clemmenson reduction (B) Wolf-Kishner reduction
 (C) Mozingo reduction (D) Bouveault-Blanc reduction
46. A mixture of acetone and CCl_4 can be separated by
- (A) Azeotropic distillation (B) Fractional distillation
 (C) Steam distillation (D) Vacuum distillation
47. In the following reaction of D-(+)glucose a product P is formed.



Amongst the following compounds, the one which will give the same product (P) under identical reaction condition is:



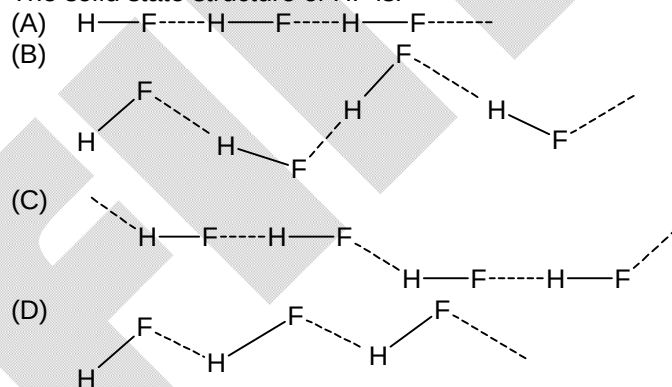
48. The correct option for the major product of the following reaction is:



49. The salt bridge in a Galvanic cell allows the flow of:

(A) ions but not electrons
 (B) both ions and electrons
 (C) electrons but not ions
 (D) neither ions nor electrons

50. The solid state structure of HF is:

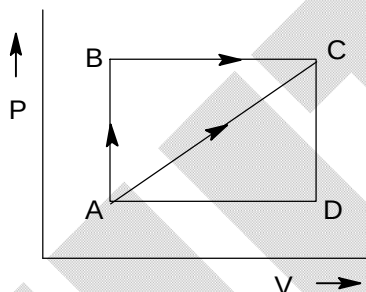


SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The number of stereoisomers for the complex $[\text{RhH}(\text{C} \equiv \text{CR})_2(\text{PMe}_3)_3]$ is:
52. The total number of lone pairs of electron on all atoms in cyanogens is 'x' and thiocyanogen is 'y'. Report $(x + y) = ?$
53. For the cell $\text{Cd} | \text{CdCl}_2 || \text{AgCl} | \text{Ag}$; $E^\circ_{\text{cell}} = 0.675 \text{ V}$ and $\left(\frac{dE^\circ}{dT}\right) = -6.5 \times 10^{-4} \text{ VK}^{-1}$ at 27°C . The $\Delta H^\circ (\text{kJ mol}^{-1})$ value for the cell is:
54. An organic compound is estimated through Dumas method and was found to evolve 264 g CO_2 , 72 g of H_2O and 28 g of N_2 . If the formula of the compound is $\text{C}_x\text{H}_y\text{N}_z$. Then $\left(\frac{x+y}{z}\right)$ is:
55. When glycinium hydrochloride is titrated against NaOH, pH at the first half equivalence point is 2.40 and the pH at the second half equivalence point is 9.60. The pH at first equivalence point is:
56. A thermodynamic process is shown below:
 Given: $P_A = 3 \times 10^4 \text{ Pa}$, $P_B = 8 \times 10^4 \text{ Pa}$, $V_A = 2 \times 10^{-3} \text{ m}^3$, $V_D = 5 \times 10^{-3} \text{ m}^3$. In the process AB, 600 J of heat is added and in BC 200 J of heat is added to the system. If the change in internal energy of the system in the process AC is x; then report $\left(\frac{x}{20}\right)$. x is calculated in Joule.



57. 10.6 g of benzaldehyde was reacted with an excess of acetophenone to produce 10.4 g of the enone product according to the given equation. Calculate the % yield of the product.
 $\text{Benzaldehyde} + \text{Acetophenone} \xrightarrow[2. \Delta]{1. \text{NaOEt}} \text{Product}$
58. Assume that the reaction of MeMgBr with ethyl acetate proceeds with 100% conversion to tert – butanol. The volume of 0.2 M solution of MeMgBr required to convert 10 mL of a 0.025 M solution of ethyl acetate is x. Report the value of $10x$ in mL.

59. The mechanism of enzyme catalysis is as follows:



The rate of product formation is given as:

$$\frac{+dP}{dt} = \frac{k_1 k_2 [E_0][S]}{k_{-1} + k_2 + k_1 [S]}, \text{ where } [E_0] \text{ is the total enzyme concentration, } [S] \text{ is substrate concentration.}$$

The rate of product formation when the substrate concentration is very large has a limiting value of 0.02 Ms^{-1} . At a substrate conc. of 250 mg/dm^3 , the rate is half this value. The value of $\frac{k_1}{k_{-1}}$ (in dm^3 / kg) assuming that $k_2 \ll k_{-1}$ is $x \times 10^3$. What is the value of x ?

60. The maximum concentration of equimolar solution of ferrous sulphate and potassium sulphide so that when mixed in equal volumes, there is no precipitate of iron sulphide is $C \times 10^{-9} \text{ M}$. Report the value of C . [Given: $K_{sp}(\text{FeS}) = 6.3 \times 10^{-18} \text{ M}^2$.

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If a non constant function $f(x) = \frac{2x^2 - kx + 5}{x^2 + 3x + 4}$ is bounded, then k can not have values is equal to
 (A) -2 (B) -6
 (C) 2 (D) 6
62. If $\sqrt{2} < \alpha, \beta < \sqrt{3}$, $\alpha \neq \beta$, then the value of $\left[\frac{\alpha - \beta}{\cot^{-1} \alpha - \cot^{-1} \beta} \right]$ is (where [.] denotes G.I.F.)
 (A) -4 (B) 4
 (C) 3 (D) -5
63. Let real number a satisfy $\{a\} + \left\{\frac{1}{a}\right\} = 1$ and b is the value of $\{a^3\} + \left\{\frac{1}{a^3}\right\}$ (where {.] denotes fractional part function), then the value of $\int_0^b \frac{x}{1+x} \sqrt{1-x^2} dx$ is
 (A) $2 + \frac{\pi}{4}$ (B) $2 - \frac{\pi}{4}$
 (C) $1 - \frac{\pi}{4}$ (D) $1 + \frac{\pi}{4}$
64. For a real number 'a' such that the equation $a(\sin x + \sin(2x)) = \sin 3x$ has more than one solution in the interval $(0, \pi)$ the set of all such value of a is
 (A) $(-3, -2)$ (B) $(-3, 1) - \{-2\}$
 (C) $(-2, 1)$ (D) $(0, 1)$
65. Let c be a real number and let z_1 and z_2 be the two complex number satisfying the equation $z^2 - cz + 10 = 0$, points $z_1, z_2, \frac{1}{z_1}$ and $\frac{1}{z_2}$ are vertices of (convex) quadrilateral in the complex plane, when the area is maximum then the value of c is
 (A) $4\sqrt{5}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{5}$ (D) $8\sqrt{5}$
66. If the equation $\frac{\alpha^2}{1 - \tan^2 x} = \frac{\sin^2 x + \alpha^2 - 2}{\cos 2x}$ has a solution, then
 (A) $\alpha^2 < 1$ (B) $\alpha^2 > 2$
 (C) $\alpha^2 \geq 1$ (D) $\alpha \in \mathbb{R}$
67. A polynomial of degree four with leading coefficient 1 and integer coefficients, has two zeros, both of which are integer. Which of the following can also be a zero of the polynomial ?
 (A) $\frac{1+i\sqrt{11}}{2}$ (B) $\frac{1+i}{2}$
 (C) $\frac{1}{2} + i$ (D) $1 + \frac{i}{2}$

68. Circle $x^2 + y^2 - 6x - 4y + 11 = 0$ is rolled up by 4 units along a tangent to it at the point (2, 3) let this be circle C_1 , C_2 is the mirror image of circle C_1 about the tangent. A and B are centres of circles C_1 and C_2 , C and D are the feet of perpendicular from A and B respectively upto x-axis, then the area of trapezium ABCD equals to
- (A) $(2 + 2\sqrt{2})$ (B) $(3 + 2\sqrt{2})$
 (C) $(6 + 4\sqrt{2})$ (D) $(4 + 4\sqrt{2})$
69. A point P is selected at random from the interior of the pentagon with vertices $A = (0, 2)$, $B = (4, 0)$, $C = (2\pi + 1, 0)$, $D = (2\pi + 1, 4)$ and $E = (0, 4)$. What is the probability that $\angle APB$ is obtuse ?
- (A) $1/5$ (B) $1/4$
 (C) $3/8$ (D) $5/16$
70. How many 15 letter arrangements of 5 A's, 5B's and 5 C's have no A's in the first 5 letter no B's in the next 5 letters and no C's in the last 5 letters
- (A) $\sum_{k=0}^5 ({}^5C_k)^3$ (B) $\sum_{k=0}^5 ({}^5P_k)^3$
 (C) 2^{15} (D) $2^5 \cdot 3^5$
71. The value of $\sum_{i=0}^n \sum_{j=1}^n {}^nC_j \cdot {}^jC_i \quad i \leq j$
- (A) 3^n (B) $3^n + 1$
 (C) $3^n - 1$ (D) 3^{n+1}
72. A parabola $y = ax^2 + bx + c$ is reflected about the x-axis, the parabola and its reflection are translated horizontally five units in opposite direction to become the graph of $y = f(x)$ and $y = g(x)$ respectively. Then which of the following describes the graph of $y = f(x) + g(x)$
- (A) A horizontal line (B) the graph of a cubic function
 (C) a non horizontal line (D) A parabola
73. If z_1 and z_2 are two complex number such that $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ where $z_1 = 4 + 6i$ and $z_2 = 3 + 5i$, then locus of z is
- (A) straight line with slope $-\frac{1}{2}$ (B) circle with radius $\frac{1}{\sqrt{2}}$
 (C) hyperbola with eccentricity $\sqrt{2}$ (D) hyperbola with eccentricity $\frac{5}{2}$
74. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$, then $\int_{1/10}^{9/10} f(f(x))dx$ is equal to
- (A) $\frac{4}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{5}$ (D) 0
75. If $I_1 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}} x^2 dx$ and $I_2 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}+1} x^2 dx$, then $\frac{I_1 - I_2}{I_2} = \frac{1}{k}$, then k is equal to
- (A) $(\sqrt{3} + 1)\sqrt{2}$ (B) $(\sqrt{3} - 1)\sqrt{2}$
 (C) $(\sqrt{3} + \sqrt{2})$ (D) $3(\sqrt{6} + \sqrt{2})$

76. Total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^{10} + 1}{t^{10} + t^4 + 1} dt = 6x - 5$ is
 (A) 0 (B) 1
 (C) 2 (D) 3
77. A continuous and differentiable function $f(x)$ satisfy the functional equation $f(x + y) = \frac{f(x)f(y)}{f(x) + f(y)}$,
 $f(1) = 32$ then $f(4) =$
 (A) 8 (B) 6
 (C) 4 (D) 2
78. $\lim_{x \rightarrow \infty} |1 - a^4|^x \sin \frac{5}{|1 - a^2|^x} a^2 > 2$ is
 (A) 0 (B) 5
 (C) ∞ (D) does not exist
79. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2 - x)$ and $f''(x) < 0 \forall x \in (0, 2)$ then which is true ?
 (A) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $(0, 2)$
 (B) $g(x)$ increases in $(0, 2)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (C) $g(x)$ increases in $\left(\frac{4}{3}, 2\right)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (D) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $\left(\frac{4}{3}, 2\right)$
80. Area bounded by the curve $y = 1$ and $y = \frac{\sin x + \cos x + |\sin x - \cos x|}{2}$ for $x \in [0, 2\pi]$ is
 (A) $2\pi - \sqrt{2}$ (B) $2\pi - 2\sqrt{2}$
 (C) $2(\pi - \sqrt{2} - 1)$ (D) $2(\pi + \sqrt{2} - 3)$

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = \sqrt{ax^2 + bx}$, for how many real values of a is there at least one positive value of b for which the domain of $f(x)$ and the range of $f(x)$ are the same set
82. Lines $x + y + 4 = 0$, $2x + ky + 2 = 0$ intersects the circle $x^2 + y^2 + 2x + 4y + 1 = 0$ and $x^2 + y^2 + 4x + 6y + 4 = 0$ as 4 points which are concyclic, then the value of k is
83. The number of distinct values of determinant of order 2×2 whose entries are from the set $\{-1, 0, 1\}$ is
84. Krishna writes the number 1, 2, 3, ..., 9 on separate cards, one number per card, he wishes to divide the cards into 3 groups of 3 cards so that the sum of the numbers in each group will be the same. In how many ways can this be done ?

85. Fifteen integers $a_1, a_2, a_3, \dots, a_{15}$ are arranged in order on a number line. The integers are equally spaced and have the property that $1 \leq a_1 \leq 10$, $13 \leq a_2 \leq 20$ and $241 \leq a_{15} \leq 250$. What is the sum of digits of a_{14} ?
86. In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term is 4000. What is the first term
87. A data set consists of 6 (not all distinct) positive integers : 1, 7, 5, 2, 5 and x, the Arithmetic Mean of the 6 numbers equals a value in the data set. What is the sum of all possible values of x
88. The area bounded by the graph of $y^2 + 2xy + 40|x| = 400$ is
89. Let the distance between two point (x_1, y_1) and (x_2, y_2) is given by $|x_1 - x_2| + |y_1 - y_2|$ for how many points P with integer coordinates is the distance between P and the origin less than or equal to 20
90. Two integers are inserted into list 3, 3, 3, 8, 11, 12, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of the two additional numbers

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2024

TEST DATE: 14-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

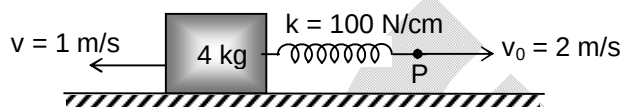
Physics

PART – A

SECTION – A (One Options Correct Type)

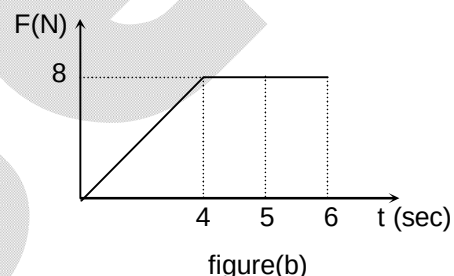
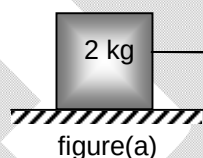
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A spring block system lies on a smooth horizontal surface. The free end of the spring is being pulled towards right with constant speed $v_0 = 2 \text{ m/s}$. At $t = 0$, the spring (spring constant $k = 100 \text{ N/cm}$) is unstretched and block is having a speed $v = 1 \text{ m/s}$ to the left. The maximum acceleration of the block during the motion is
- (A) 50 m/s^2
(C) 150 m/s^2



- (B) 100 m/s^2
(D) 200 m/s^2

2. A block of mass 2 kg is kept on a rough horizontal surface having coefficient of friction ($\mu_s = 0.3$ and $\mu_k = 0.25$). A time dependent force F is applied on the block as shown in the figure (b). Find the speed of the block at $t = 5 \text{ sec}$.



- (A) 12 m/s
(C) 2.5 m/s

- (B) zero
(D) 0.5 m/s

3. A container of fixed volume has mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T . Assuming the gases are ideal, then choose the INCORRECT Statement.
- (A) Internal energy of the mixture is $4RT$ Joule

- (B) Ratio of rms speed of gas molecules in the mixture and hydrogen is $\sqrt{\frac{2}{3}}$

- (C) Ratio of speed of sound in the mixture and helium is $\sqrt{\frac{6}{5}}$

- (D) Ratio of rms speed of the gas molecules in Helium and hydrogen is $\sqrt{2}$

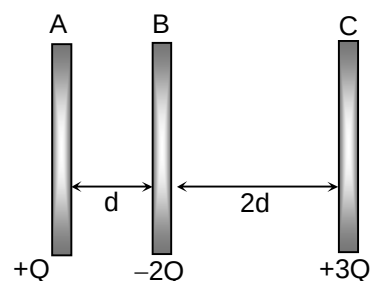
4. Three large metallic parallel plates A, B and C each having area A are separated by very small distances as shown in the figure. Charges Q , $-2Q$ and $3Q$ are given to the plates. The potential difference between plates C and A is

- (A) zero

- (B) $\frac{2Qd}{\epsilon_0 A}$

- (C) $\frac{4Qd}{\epsilon_0 A}$

- (D) Indeterminate

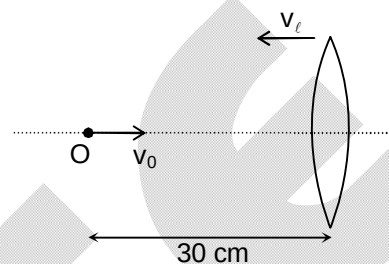


5. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$ and comes out without any change in magnitude or direction of $\vec{v}$. Then (here $\vec{E}$ is electric field and $\vec{B}$ is magnetic field)

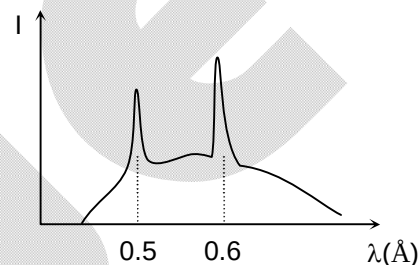
(A) $\vec{B} \times \vec{E} = |\vec{E}|^2 \vec{v}$ (B) $\vec{E} \times \vec{B} = |\vec{B}|^2 \vec{v}$
 (C) $\vec{B} \times \vec{E} = |\vec{B}|^2 \vec{v}$ (D) $\vec{E} \times \vec{B} = |\vec{E}|^2 \vec{v}$

6. A point object has velocity $v_0 = 2$ cm/s when it is placed at a distance of 30 cm from a lens moving with a speed $v_\ell = 2$ cm/s towards left. If the focal length of the lens is $f = 20$ cm, find the velocity of the image with respect to the ground.

- (A) 14 cm/s towards right
 (B) 14 cm/s towards left
 (C) 25 cm/s towards right
 (D) 25 cm/s towards left



7. The graph shows the intensity of X-rays emitted by a heavy target and the wavelength, when electrons of energy 20 keV are incident on the target. In the graph one peak is of K_α line and the other peak is of K_β line. Then



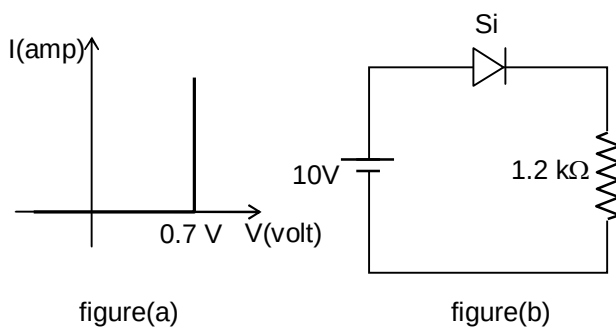
- (A) If the energy of incident electron is increased to 30 keV, then both the peaks will shift rightward.
 (B) If the energy of incident electron is increased to 30 keV, then both the peaks will shift leftward.
 (C) K_α line wavelength is 0.5 Å
 (D) K_α line wavelength is 0.6 Å

8. A 2 kg particle is moving along the x-axis under the action of the force $\vec{F} = -\frac{\pi^2}{32} \vec{x}$, where F is in newton and x is in meter. At $t = 2$ sec, the particle is moving along $\vec{x}$ and is at $x = 0$. At $t = 10$ sec it is observed that particle's speed is $4\sqrt{2}$ m/s. Then amplitude of particle motion is

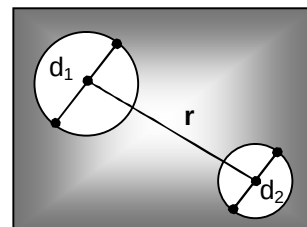
- (A) $\frac{32}{\pi}$ m (B) $\frac{32\sqrt{2}}{\pi}$ m
 (C) $\frac{16\sqrt{2}}{\pi}$ m (D) $\frac{16}{\pi}$ m

9. In the figure (a) I-V characteristic of Si-diode is shown. In figure(b) shows a circuit having the Si-diode. The current in the circuit is

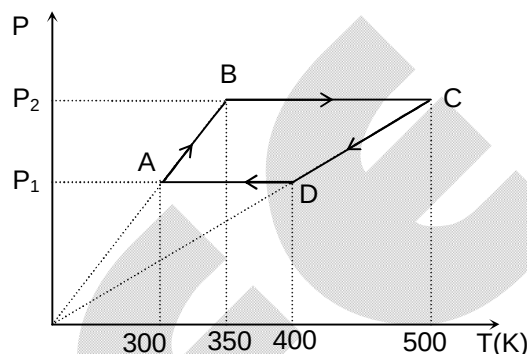
- (A) 8.33 mA
 (B) 0
 (C) 8.92 mA
 (D) 7.75 mA



10. Two holes of unequal diameters d_1 and d_2 ($d_2 < d_1$) cut in metal sheet. Separation between the centres of the holes is r . Now the sheet is heated to a higher temperature. Then
- (A) d_1, d_2 increases and r decreases
 (B) d_1, d_2 increases and r increases
 (C) d_1, d_2 decreases and r increases
 (D) d_1, d_2 decreases and r decreases



11. P-T curve of a cyclic process is shown. If number of moles of the monoatomic ideal gas are n . Then efficiency of the cycle is
- (A) 1
 (B) $1/3$
 (C) $1/9$
 (D) $1/11$

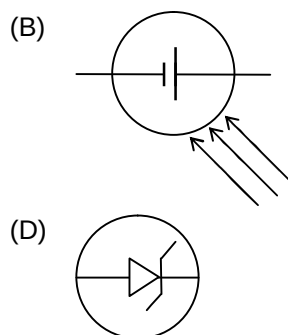
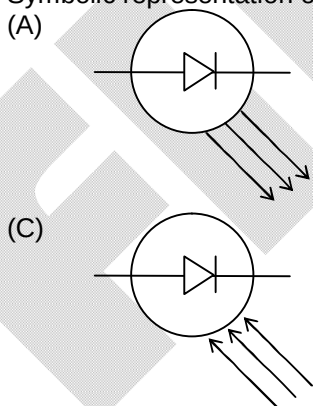


12. In an xyz space only electric field is present having potential function $V = k(x^2 + y^2 + z^2)$ volt, where k is a positive constant and x, y, z are co-ordinate of a point in meter unit. A positive charge particle of charge q and mass m is placed at $(1m, 1m, 1m)$. Acceleration of the particle at the point is

- (A) $\frac{6kq}{m}$
 (B) $\frac{\sqrt{3}kq}{m}$
 (C) $\frac{\sqrt{3}kq}{2m}$
 (D) $\frac{2\sqrt{3}kq}{m}$

13. A nucleus X of mass number 218 in free state decays to emit an α -particle. Kinetic energy of the α -particle emitted is 6.7 MeV. The Q-value of the α -decay nuclear reaction is (in MeV)
- (A) 0.125
 (B) 6.825
 (C) 6.8
 (D) 6.7

14. Symbolic representation of photodiode is

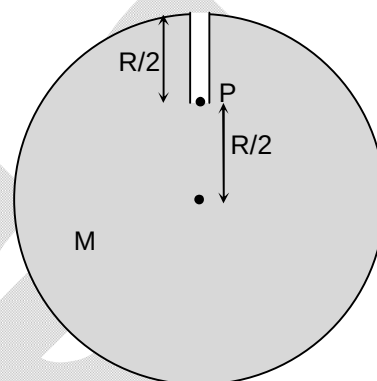


15. Two coherent sound sources in phase of frequencies of 256 Hz and 260 Hz are sounding together. If an observer records an intensity of I_0 at $t = 0$, then he will observe intensity at $t = \frac{1}{16}$ sec.

(A) I_0 (B) $\frac{I_0}{2}$
(C) $\frac{I_0}{4}$ (D) zero

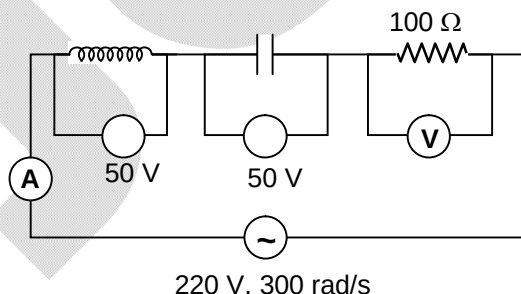
16. A particle is projected from a depth $\frac{R}{2}$ inside the earth such that it escapes to infinity. Mass of the particle is negligibly small is compared to mass of earth M . Then escape speed from the point P is

(A) $\sqrt{\frac{2GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$
(C) $\sqrt{\frac{13GM}{4R}}$ (D) $\sqrt{\frac{11GM}{4R}}$



17. For the AC circuit in the figure, choose the correct option having reading of AC voltmeter (V) and AC ammeter (A).

(A) 120 V, 1.2 A
(B) 220 V, 2.2 A
(C) 320 V, 3.2 A
(D) 220V, 1.2 A



18. The color code of a resistance is given in the figure. The value of resistance and tolerance, respectively are

(A) 47 kΩ, 10%
(B) 4.7 kΩ, 5%
(C) 470 Ω, 5%
(D) 470 kΩ, 5%

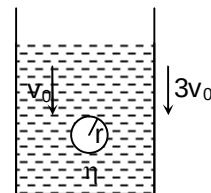


19. A non conducting ring of mass m and radius R a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that the plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0 t^2$ tesla is switched on at $t = 0$. After 2 second for switching on the magnetic field the ring is just about to rotate about vertical axis through its centre. m , R , Q , B are in SI unit. Then

(A) The induced electric field is steady with respect to time in the space
(B) the net electric force acting on the ring at any time t is $QB_R t$
(C) Friction first acts at $t = 2$ sec

(D) The coefficient of static friction between the ring and the surface is $\frac{2B_0 R Q}{mg}$

20. A container filled with viscous liquid is moving vertically downward with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertical downwards in liquid has a speed v_0 with respect to ground. The coefficient of viscosity is η . There is no relative motion between the container and the liquid. Then at the shown instant, the power developed by viscous force with respect to ground frame is



- (A) $-6\pi\eta rv_0^2$ (B) zero
(C) $12\pi\eta rv_0^2$ (D) $-12\pi\eta rv_0^2$

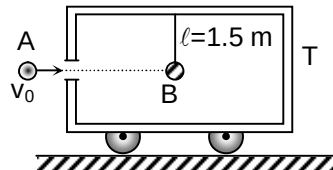
SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

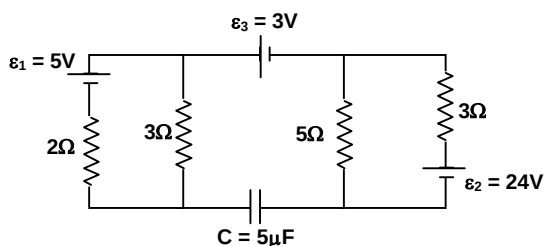
21. An unpolarized light of intensity 80 W/m^2 passes through three polarizing sheets whose pass axes make an angle of 30° , 60° and 30° with y-axis in same sense. The final transmitted intensity after passing through all three polarizers is $I \text{ W/m}^2$. Then find the value of $2I$.
22. A particle of specific charge $2 \times 10^{-2} \text{ C/kg}$ starts moving from the origin under the action of a uniform electric field $\vec{E} = (100 \text{ N/C})\hat{i}$ and magnetic field $\vec{B} = (5\text{T})\hat{k}$. Its velocity vector at $(a, b, 0) \text{ m}$ is $(4\hat{i} - 3\hat{j}) \text{ m/s}$. Then value of $4a$ is (in m)
23. A object is projected with a velocity of 10 m/s making an angle of 30° with horizontal from origin of x-y vertical plane having positive x-axis along horizontal and positive y-axis along the vertically upward direction. The equation of the trajectory is $y = Ax - Bx^2$ where x and y are co-ordinates at a point in meter. Find the value of $\sqrt{3} \left(\frac{A}{B} \right)$ (in SI unit). (Take $g = 10 \text{ m/s}^2$)
24. Each side of a cube is measured to be 7.203 m . If the total surface area of the cube considering the correct significant digits in m^2 is A , then find the value of $10A$.
25. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in 40 seconds is $\tau \text{ N-m}$. Find the value of $\pi\tau$. (Take $\pi^2 = 10$)

26. In the figure shown a ball A of mass $m = 1 \text{ kg}$ collide with another identical ball B with a speed $v_0 = 6 \text{ m/s}$ and sticks to it. The ball B is hanging inside a trolley T of mass $M = 4 \text{ kg}$. Find the maximum height (in cm) attained by the ball (A + B) from its initial level after the collision (Take $g = 10 \text{ m/s}^2$)



27. The solar constant (intensity of solar radiation) just at Venus surface measured to be 2600 W/m^2 . If the radius of the Sun is $7 \times 10^8 \text{ m}$, while the Venus-Sun distance is $1.1 \times 10^{11} \text{ m}$, then find the most probable wavelength in the solar radiation in nm. [Take $(1.81)^{\frac{1}{4}} = 1.16$, wein's constant $b = 2.9 \times 10^{-3} \text{ m-K}$] Sun can be assumed as a black body.

28. If energy stored in the capacitor C in the steady state is $U_0 \mu\text{J}$, then find $2U_0$.



29. A parallel plate capacitor with $A = 1.2 \text{ m}^2$ and separation $d = 1 \text{ mm}$ is charged by a current $i = (8\text{A})e^{-\frac{t}{4\text{sec}}}$. Consider a plane surface X of area 0.6 m^2 parallel to the plates and drawn symmetrically between the plates. Find the displacement current (in mA) through the plane surface X at $t = 4 \text{ sec}$. (Take $e^{-1} = 0.37$)
30. A submarine A travelling at 18 km/hr is being chased along the line of its velocity by another submarine B travelling at 27 km/hr . B send a Sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν (in Hz) is close to (take speed of sound in water = 1500 m/s)

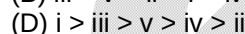
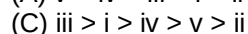
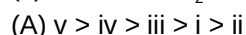
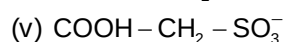
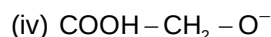
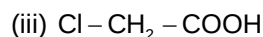
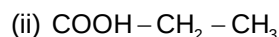
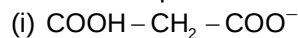
Chemistry

PART – B

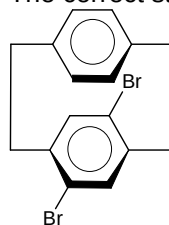
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Compare the acidic strength of following compounds and arrange in decreasing order. Choose the correct option



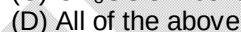
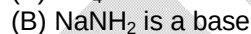
32. The correct statement about the following molecule is:



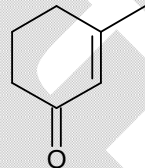
- (A) Molecule is chiral and does not possess chiral centre.
 (B) Molecule is chiral and possess chiral centre.
 (C) Molecule is achiral due to plane of symmetry.
 (D) Molecule is achiral due to axis of symmetry.

33. Choose the correct statement:

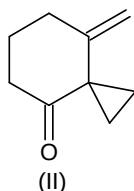
In liquid ammonia:



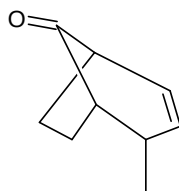
34. The product(s) are :



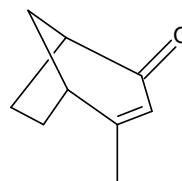
(I)



(II)



(III)



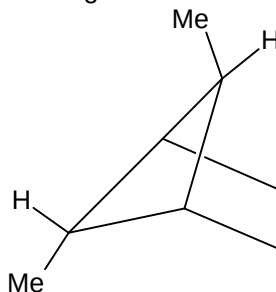
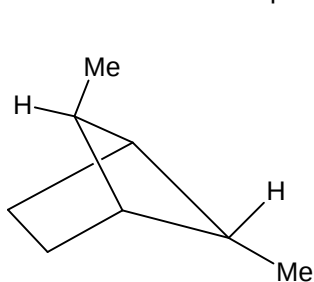
(IV)

- (A) I and III
 (C) I and II

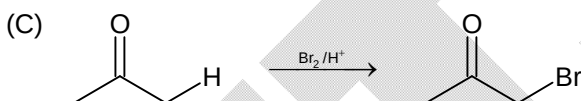
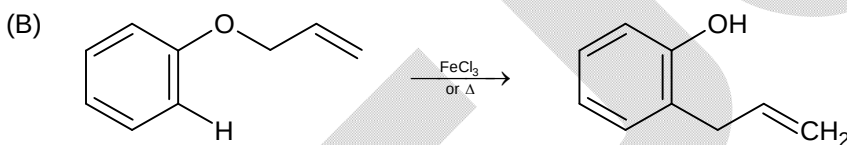
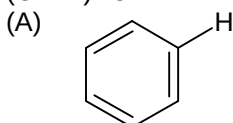
- (B) II and IV
 (D) II and III

35. Considering N as central atom, the structure of $(\text{CH}_3)_2\text{NCS}$ and $(\text{SiH}_3)_2\text{NCS}$ respectively, are:
 (A) Linear and Bent (B) Bent and Bent
 (C) Linear and Linear (D) Bent and linear

36. The correct relationship between the following structures is that they are:



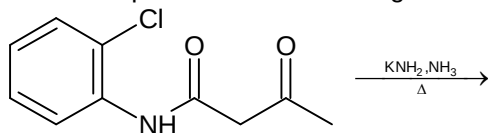
- (A) Enantiomers (B) Identical
 (C) Diastereomers (D) Constitutional isomers
37. The reaction that is expected to show primary kinetic isotope effect for the indicated H-atom (C – H) is:



38. The pair in which both actinides show +3 oxidation state only is:
 (A) Ac and Lr (B) Ac and No
 (C) Cm and Bk (D) Cm and Lr
39. I_2 is violet in the solid as well as in gas phase. However, in acetone or ethanol, it turns brown. Choose the correct statement(s) for this colour change:
 (i) Dissociation of I_2 in atomic state.
 (ii) Interaction of low-lying σ^* orbital of iodine with lone pair of O (solvent).
 (iii) Formation of a charge transfer complex.
 (A) (i) only (B) (ii) only
 (C) (i) and (ii) only (D) (iii) only

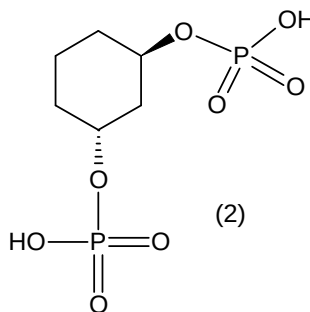
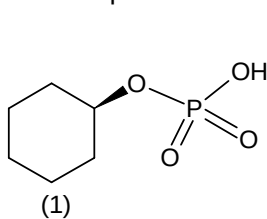
40. An atomic orbital is given by $nr^2 e^{-\frac{r}{3a_0}} (3\cos^2\theta - 1)$ in hydrogen atom. The orbital represents:
 (A) 2p (B) 3p
 (C) 3d (D) 4d

41. The magnitude of bond angles in gaseous NF_3 , SbF_3 and SbCl_3 follows the order:
 (A) $\text{NF}_3 > \text{NH}_3 > \text{SbCl}_3$ (B) $\text{SbCl}_3 > \text{NH}_3 > \text{NF}_3$
 (C) $\text{NH}_3 > \text{NF}_3 > \text{SbCl}_3$ (D) $\text{NF}_3 > \text{SbCl}_3 > \text{NH}_3$
42. Consider the following chemical reactions in gas phase
 Reaction I : $\text{H} + \text{D}_2 \longrightarrow \text{HD} + \text{D}$
 Reaction II : $\text{D} + \text{H}_2 \longrightarrow \text{HD} + \text{H}$
 Which of the following statement is true?
 (A) Reaction I and II, both are endothermic.
 (B) Reaction I is endothermic, Reaction II is exothermic.
 (C) Both reactions are thermo neutral as they involve isotopes which differ by the number of neutrons, which do not play any role in chemical reactions.
 (D) Reaction I and II both are exothermic.
43. Predict the product of the following reaction

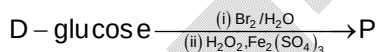


- (A)
- (B)
- (C)
- (D)

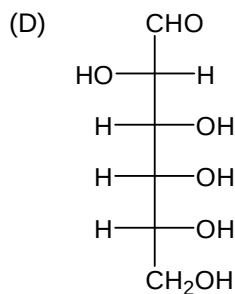
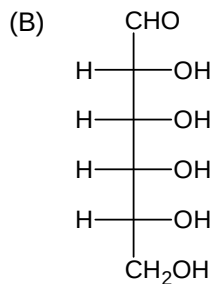
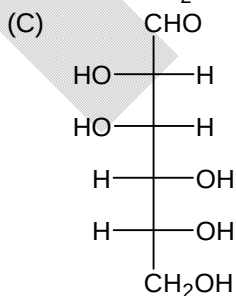
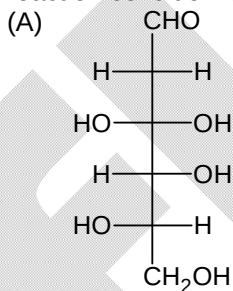
44. Compound 2 and compound 1 were mixed in a 1 : 4 molar ratio. The mixture was subjected to an ammonium molybdate test for phosphate estimation. The result of the molybdated test yielded a phosphate concentration of 36.0 mM for the mixture. What are the concentration of compound 1 and compound 2 in the mixture?



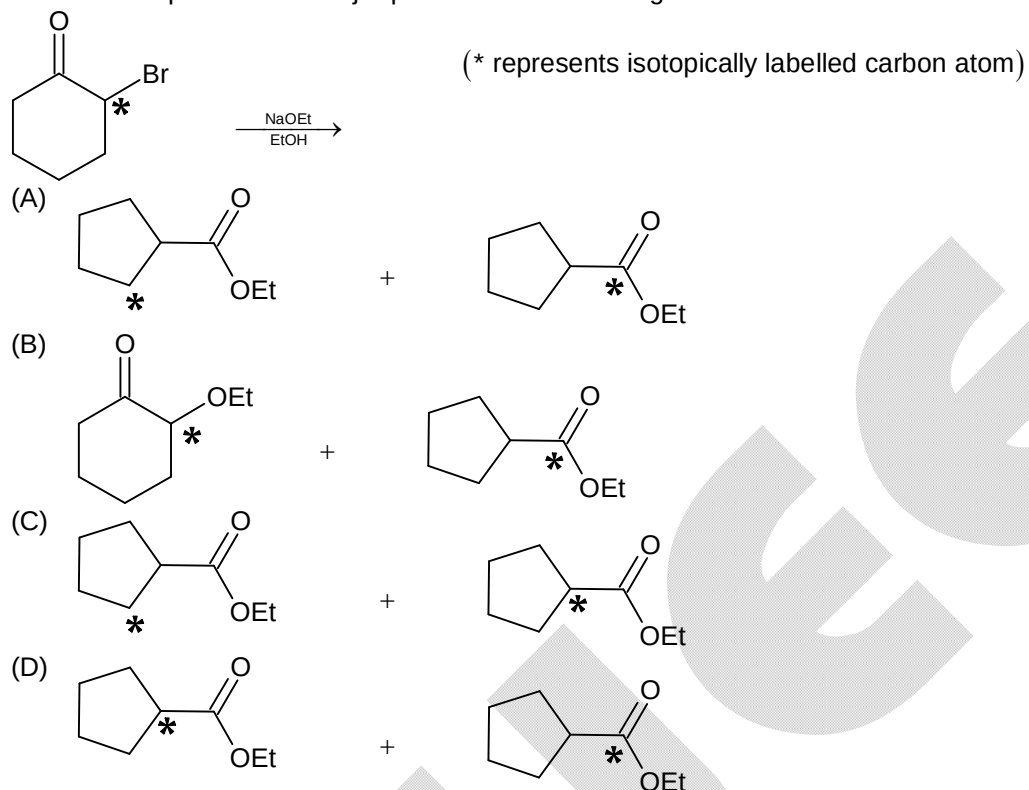
- (A) Compound 1 : 30 mM Compound 2 : 6 mM
 (B) Compound 1 : 24 mM Compound 2 : 6 mM
 (C) Compound 1 : 28.8 mM Compound 2 : 7.2 mM
 (D) Compound 1 : 24 mM Compound 2 : 12 mM
45. Which of the following can't be used for the given transformation?
- $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{C}(\text{CH}_3)_3 \longrightarrow \text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)_3$
- (A) Clemmenson reduction (B) Wolf-Kishner reduction
 (C) Mozingo reduction (D) Bouveault-Blanc reduction
46. A mixture of acetone and CCl_4 can be separated by
- (A) Azeotropic distillation (B) Fractional distillation
 (C) Steam distillation (D) Vacuum distillation
47. In the following reaction of D-(+)glucose a product P is formed.



Amongst the following compounds, the one which will give the same product (P) under identical reaction condition is:



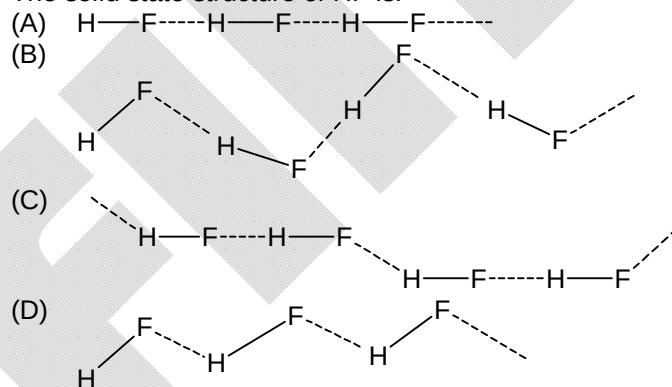
48. The correct option for the major product of the following reaction is:



49. The salt bridge in a Galvanic cell allows the flow of:

(A) ions but not electrons
 (B) both ions and electrons
 (C) electrons but not ions
 (D) neither ions nor electrons

50. The solid state structure of HF is:

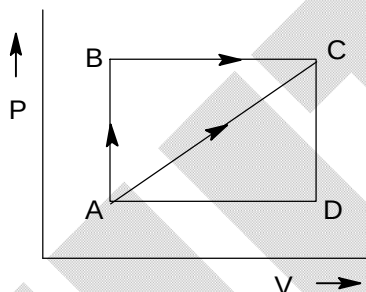


SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The number of stereoisomers for the complex $[\text{RhH}(\text{C} \equiv \text{CR})_2(\text{PMe}_3)_3]$ is:
52. The total number of lone pairs of electron on all atoms in cyanogens is 'x' and thiocyanogen is 'y'. Report $(x + y) = ?$
53. For the cell $\text{Cd} | \text{CdCl}_2 || \text{AgCl} | \text{Ag}$; $E^\circ_{\text{cell}} = 0.675 \text{ V}$ and $\left(\frac{dE^\circ}{dT}\right) = -6.5 \times 10^{-4} \text{ VK}^{-1}$ at 27°C . The $\Delta H^\circ (\text{kJ mol}^{-1})$ value for the cell is:
54. An organic compound is estimated through Dumas method and was found to evolve 264 g CO_2 , 72 g of H_2O and 28 g of N_2 . If the formula of the compound is $\text{C}_x\text{H}_y\text{N}_z$. Then $\left(\frac{x+y}{z}\right)$ is:
55. When glycinium hydrochloride is titrated against NaOH , pH at the first half equivalence point is 2.40 and the pH at the second half equivalence point is 9.60. The pH at first equivalence point is:
56. A thermodynamic process is shown below:
Given: $P_A = 3 \times 10^4 \text{ Pa}$, $P_B = 8 \times 10^4 \text{ Pa}$, $V_A = 2 \times 10^{-3} \text{ m}^3$, $V_D = 5 \times 10^{-3} \text{ m}^3$. In the process AB, 600 J of heat is added and in BC 200 J of heat is added to the system. If the change in internal energy of the system in the process AC is x; then report $\left(\frac{x}{20}\right)$. x is calculated in Joule.



57. 10.6 g of benzaldehyde was reacted with an excess of acetophenone to produce 10.4 g of the enone product according to the given equation. Calculate the % yield of the product.
Benzaldehyde + Acetophenone $\xrightarrow[2. \Delta]{1. \text{NaOEt}}$ Product
58. Assume that the reaction of MeMgBr with ethyl acetate proceeds with 100% conversion to tert – butanol. The volume of 0.2 M solution of MeMgBr required to convert 10 mL of a 0.025 M solution of ethyl acetate is x. Report the value of $10x$ in mL.

59. The mechanism of enzyme catalysis is as follows:



The rate of product formation is given as:

$$\frac{+dP}{dt} = \frac{k_1 k_2 [E_0][S]}{k_{-1} + k_2 + k_1 [S]}, \text{ where } [E_0] \text{ is the total enzyme concentration, } [S] \text{ is substrate concentration.}$$

The rate of product formation when the substrate concentration is very large has a limiting value of 0.02 Ms^{-1} . At a substrate conc. of 250 mg/dm^3 , the rate is half this value. The value of $\frac{k_1}{k_{-1}}$ (in dm^3 / kg) assuming that $k_2 \ll k_{-1}$ is $x \times 10^3$. What is the value of x ?

60. The maximum concentration of equimolar solution of ferrous sulphate and potassium sulphide so that when mixed in equal volumes, there is no precipitate of iron sulphide is $C \times 10^{-9} \text{ M}$. Report the value of C . [Given: $K_{sp}(\text{FeS}) = 6.3 \times 10^{-18} \text{ M}^2$.

Mathematics**PART – C****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If a non constant function $f(x) = \frac{2x^2 - kx + 5}{x^2 + 3x + 4}$ is bounded, then k can not have values is equal to
 (A) -2 (B) -6
 (C) 2 (D) 6
62. If $\sqrt{2} < \alpha, \beta < \sqrt{3}$, $\alpha \neq \beta$, then the value of $\left[\frac{\alpha - \beta}{\cot^{-1} \alpha - \cot^{-1} \beta} \right]$ is (where [.] denotes G.I.F.)
 (A) -4 (B) 4
 (C) 3 (D) -5
63. Let real number a satisfy $\{a\} + \left\{\frac{1}{a}\right\} = 1$ and b is the value of $\{a^3\} + \left\{\frac{1}{a^3}\right\}$ (where {.] denotes fractional part function), then the value of $\int_0^b \frac{x}{1+x} \sqrt{1-x^2} dx$ is
 (A) $2 + \frac{\pi}{4}$ (B) $2 - \frac{\pi}{4}$
 (C) $1 - \frac{\pi}{4}$ (D) $1 + \frac{\pi}{4}$
64. For a real number 'a' such that the equation $a(\sin x + \sin(2x)) = \sin 3x$ has more than one solution in the interval $(0, \pi)$ the set of all such value of a is
 (A) $(-3, -2)$ (B) $(-3, 1) - \{-2\}$
 (C) $(-2, 1)$ (D) $(0, 1)$
65. Let c be a real number and let z_1 and z_2 be the two complex number satisfying the equation $z^2 - cz + 10 = 0$, points $z_1, z_2, \frac{1}{z_1}$ and $\frac{1}{z_2}$ are vertices of (convex) quadrilateral in the complex plane, when the area is maximum then the value of c is
 (A) $4\sqrt{5}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{5}$ (D) $8\sqrt{5}$
66. If the equation $\frac{\alpha^2}{1 - \tan^2 x} = \frac{\sin^2 x + \alpha^2 - 2}{\cos 2x}$ has a solution, then
 (A) $\alpha^2 < 1$ (B) $\alpha^2 > 2$
 (C) $\alpha^2 \geq 1$ (D) $\alpha \in \mathbb{R}$
67. A polynomial of degree four with leading coefficient 1 and integer coefficients, has two zeros, both of which are integer. Which of the following can also be a zero of the polynomial ?
 (A) $\frac{1+i\sqrt{11}}{2}$ (B) $\frac{1+i}{2}$
 (C) $\frac{1}{2} + i$ (D) $1 + \frac{i}{2}$

68. Circle $x^2 + y^2 - 6x - 4y + 11 = 0$ is rolled up by 4 units along a tangent to it at the point (2, 3) let this be circle C_1 , C_2 is the mirror image of circle C_1 about the tangent. A and B are centres of circles C_1 and C_2 , C and D are the feet of perpendicular from A and B respectively upto x-axis, then the area of trapezium ABCD equals to
- (A) $(2 + 2\sqrt{2})$ (B) $(3 + 2\sqrt{2})$
 (C) $(6 + 4\sqrt{2})$ (D) $(4 + 4\sqrt{2})$
69. A point P is selected at random from the interior of the pentagon with vertices $A = (0, 2)$, $B = (4, 0)$, $C = (2\pi + 1, 0)$, $D = (2\pi + 1, 4)$ and $E = (0, 4)$. What is the probability that $\angle APB$ is obtuse ?
- (A) $1/5$ (B) $1/4$
 (C) $3/8$ (D) $5/16$
70. How many 15 letter arrangements of 5 A's, 5B's and 5 C's have no A's in the first 5 letter no B's in the next 5 letters and no C's in the last 5 letters
- (A) $\sum_{k=0}^5 ({}^5C_k)^3$ (B) $\sum_{k=0}^5 ({}^5P_k)^3$
 (C) 2^{15} (D) $2^5 \cdot 3^5$
71. The value of $\sum_{i=0}^n \sum_{j=1}^n {}^nC_j \cdot {}^jC_i \quad i \leq j$
- (A) 3^n (B) $3^n + 1$
 (C) $3^n - 1$ (D) 3^{n+1}
72. A parabola $y = ax^2 + bx + c$ is reflected about the x-axis, the parabola and its reflection are translated horizontally five units in opposite direction to become the graph of $y = f(x)$ and $y = g(x)$ respectively. Then which of the following describes the graph of $y = f(x) + g(x)$
- (A) A horizontal line (B) the graph of a cubic function
 (C) a non horizontal line (D) A parabola
73. If z_1 and z_2 are two complex number such that $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ where $z_1 = 4 + 6i$ and $z_2 = 3 + 5i$, then locus of z is
- (A) straight line with slope $-\frac{1}{2}$ (B) circle with radius $\frac{1}{\sqrt{2}}$
 (C) hyperbola with eccentricity $\sqrt{2}$ (D) hyperbola with eccentricity $\frac{5}{2}$
74. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$, then $\int_{1/10}^{9/10} f(f(x))dx$ is equal to
- (A) $\frac{4}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{5}$ (D) 0
75. If $I_1 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}} x^2 dx$ and $I_2 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}+1} x^2 dx$, then $\frac{I_1 - I_2}{I_2} = \frac{1}{k}$, then k is equal to
- (A) $(\sqrt{3} + 1)\sqrt{2}$ (B) $(\sqrt{3} - 1)\sqrt{2}$
 (C) $(\sqrt{3} + \sqrt{2})$ (D) $3(\sqrt{6} + \sqrt{2})$

76. Total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^{10} + 1}{t^{10} + t^4 + 1} dt = 6x - 5$ is
 (A) 0 (B) 1
 (C) 2 (D) 3
77. A continuous and differentiable function $f(x)$ satisfy the functional equation $f(x + y) = \frac{f(x)f(y)}{f(x) + f(y)}$,
 $f(1) = 32$ then $f(4) =$
 (A) 8 (B) 6
 (C) 4 (D) 2
78. $\lim_{x \rightarrow \infty} |1 - a^4|^x \sin \frac{5}{|1 - a^2|^x} a^2 > 2$ is
 (A) 0 (B) 5
 (C) ∞ (D) does not exist
79. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2 - x)$ and $f''(x) < 0 \forall x \in (0, 2)$ then which is true ?
 (A) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $(0, 2)$
 (B) $g(x)$ increases in $(0, 2)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (C) $g(x)$ increases in $\left(\frac{4}{3}, 2\right)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (D) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $\left(\frac{4}{3}, 2\right)$
80. Area bounded by the curve $y = 1$ and $y = \frac{\sin x + \cos x + |\sin x - \cos x|}{2}$ for $x \in [0, 2\pi]$ is
 (A) $2\pi - \sqrt{2}$ (B) $2\pi - 2\sqrt{2}$
 (C) $2(\pi - \sqrt{2} - 1)$ (D) $2(\pi + \sqrt{2} - 3)$

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = \sqrt{ax^2 + bx}$, for how many real values of a is there at least one positive value of b for which the domain of $f(x)$ and the range of $f(x)$ are the same set
82. Lines $x + y + 4 = 0$, $2x + ky + 2 = 0$ intersects the circle $x^2 + y^2 + 2x + 4y + 1 = 0$ and $x^2 + y^2 + 4x + 6y + 4 = 0$ as 4 points which are concyclic, then the value of k is
83. The number of distinct values of determinant of order 2×2 whose entries are from the set $\{-1, 0, 1\}$ is
84. Krishna writes the number 1, 2, 3, ..., 9 on separate cards, one number per card, he wishes to divide the cards into 3 groups of 3 cards so that the sum of the numbers in each group will be the same. In how many ways can this be done ?

85. Fifteen integers $a_1, a_2, a_3, \dots, a_{15}$ are arranged in order on a number line. The integers are equally spaced and have the property that $1 \leq a_1 \leq 10$, $13 \leq a_2 \leq 20$ and $241 \leq a_{15} \leq 250$. What is the sum of digits of a_{14} ?
86. In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term is 4000. What is the first term
87. A data set consists of 6 (not all distinct) positive integers : 1, 7, 5, 2, 5 and x, the Arithmetic Mean of the 6 numbers equals a value in the data set. What is the sum of all possible values of x
88. The area bounded by the graph of $y^2 + 2xy + 40|x| = 400$ is
89. Let the distance between two point (x_1, y_1) and (x_2, y_2) is given by $|x_1 - x_2| + |y_1 - y_2|$ for how many points P with integer coordinates is the distance between P and the origin less than or equal to 20
90. Two integers are inserted into list 3, 3, 3, 8, 11, 12, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of the two additional numbers

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2024

TEST DATE: 14-01-2024

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

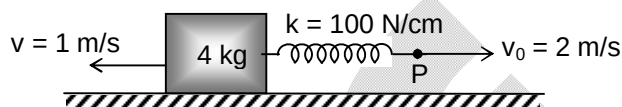
Physics

PART – A

SECTION – A (One Options Correct Type)

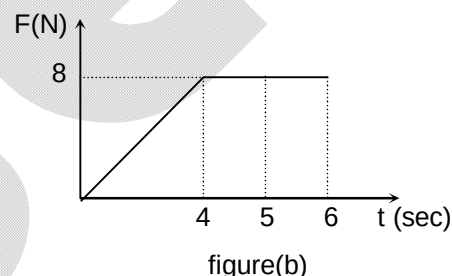
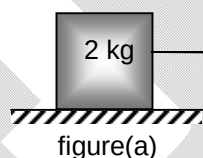
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A spring block system lies on a smooth horizontal surface. The free end of the spring is being pulled towards right with constant speed $v_0 = 2 \text{ m/s}$. At $t = 0$, the spring (spring constant $k = 100 \text{ N/cm}$) is unstretched and block is having a speed $v = 1 \text{ m/s}$ to the left. The maximum acceleration of the block during the motion is
- (A) 50 m/s^2
(C) 150 m/s^2



- (B) 100 m/s^2
(D) 200 m/s^2

2. A block of mass 2 kg is kept on a rough horizontal surface having coefficient of friction ($\mu_s = 0.3$ and $\mu_k = 0.25$). A time dependent force F is applied on the block as shown in the figure (b). Find the speed of the block at $t = 5 \text{ sec}$.



- (A) 12 m/s
(C) 2.5 m/s

- (B) zero
(D) 0.5 m/s

3. A container of fixed volume has mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T . Assuming the gases are ideal, then choose the INCORRECT Statement.
- (A) Internal energy of the mixture is $4RT$ Joule

- (B) Ratio of rms speed of gas molecules in the mixture and hydrogen is $\sqrt{\frac{2}{3}}$

- (C) Ratio of speed of sound in the mixture and helium is $\sqrt{\frac{6}{5}}$

- (D) Ratio of rms speed of the gas molecules in Helium and hydrogen is $\sqrt{2}$

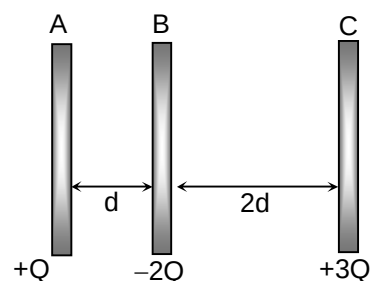
4. Three large metallic parallel plates A, B and C each having area A are separated by very small distances as shown in the figure. Charges Q , $-2Q$ and $3Q$ are given to the plates. The potential difference between plates C and A is

- (A) zero

- (B) $\frac{2Qd}{\epsilon_0 A}$

- (C) $\frac{4Qd}{\epsilon_0 A}$

- (D) Indeterminate

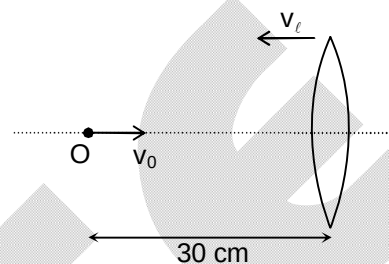


5. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$ and comes out without any change in magnitude or direction of $\vec{v}$. Then (here $\vec{E}$ is electric field and $\vec{B}$ is magnetic field)

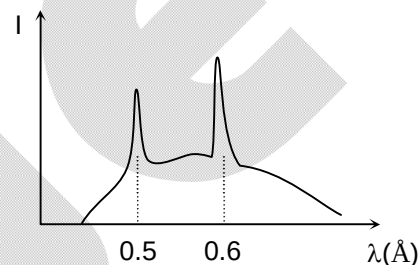
(A) $\vec{B} \times \vec{E} = |\vec{E}|^2 \vec{v}$ (B) $\vec{E} \times \vec{B} = |\vec{B}|^2 \vec{v}$
 (C) $\vec{B} \times \vec{E} = |\vec{B}|^2 \vec{v}$ (D) $\vec{E} \times \vec{B} = |\vec{E}|^2 \vec{v}$

6. A point object has velocity $v_0 = 2$ cm/s when it is placed at a distance of 30 cm from a lens moving with a speed $v_\ell = 2$ cm/s towards left. If the focal length of the lens is $f = 20$ cm, find the velocity of the image with respect to the ground.

- (A) 14 cm/s towards right
 (B) 14 cm/s towards left
 (C) 25 cm/s towards right
 (D) 25 cm/s towards left



7. The graph shows the intensity of X-rays emitted by a heavy target and the wavelength, when electrons of energy 20 keV are incident on the target. In the graph one peak is of K_α line and the other peak is of K_β line. Then



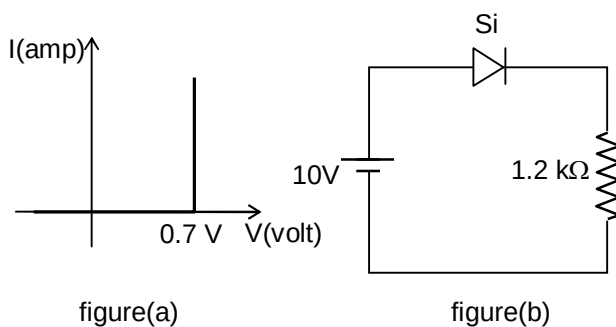
- (A) If the energy of incident electron is increased to 30 keV, then both the peaks will shift rightward.
 (B) If the energy of incident electron is increased to 30 keV, then both the peaks will shift leftward.
 (C) K_α line wavelength is 0.5 Å
 (D) K_α line wavelength is 0.6 Å

8. A 2 kg particle is moving along the x-axis under the action of the force $\vec{F} = -\frac{\pi^2}{32} \vec{x}$, where F is in newton and x is in meter. At $t = 2$ sec, the particle is moving along $\vec{x}$ and is at $x = 0$. At $t = 10$ sec it is observed that particle's speed is $4\sqrt{2}$ m/s. Then amplitude of particle motion is

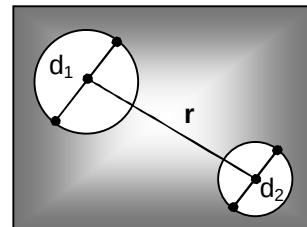
(A) $\frac{32}{\pi}$ m (B) $\frac{32\sqrt{2}}{\pi}$ m
 (C) $\frac{16\sqrt{2}}{\pi}$ m (D) $\frac{16}{\pi}$ m

9. In the figure (a) I-V characteristic of Si-diode is shown. In figure(b) shows a circuit having the Si-diode. The current in the circuit is

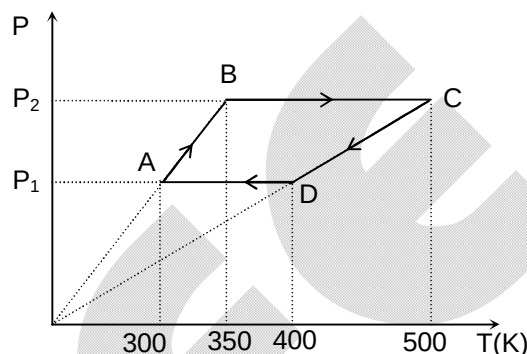
- (A) 8.33 mA
 (B) 0
 (C) 8.92 mA
 (D) 7.75 mA



10. Two holes of unequal diameters d_1 and d_2 ($d_2 < d_1$) cut in metal sheet. Separation between the centres of the holes is r . Now the sheet is heated to a higher temperature. Then
- (A) d_1, d_2 increases and r decreases
 (B) d_1, d_2 increases and r increases
 (C) d_1, d_2 decreases and r increases
 (D) d_1, d_2 decreases and r decreases



11. P-T curve of a cyclic process is shown. If number of moles of the monoatomic ideal gas are n . Then efficiency of the cycle is
- (A) 1
 (B) $1/3$
 (C) $1/9$
 (D) $1/11$

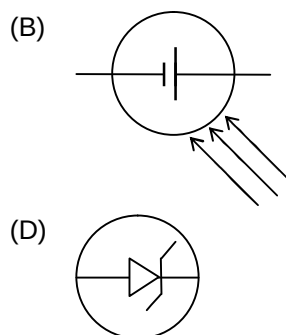
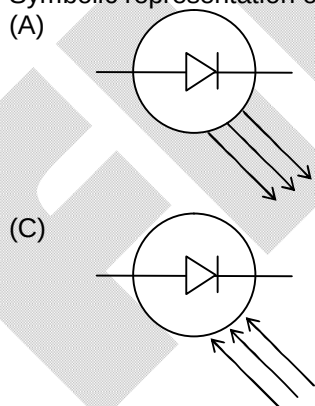


12. In an xyz space only electric field is present having potential function $V = k(x^2 + y^2 + z^2)$ volt, where k is a positive constant and x, y, z are co-ordinate of a point in meter unit. A positive charge particle of charge q and m is placed at $(1m, 1m, 1m)$. Acceleration of the particle at the point is

- (A) $\frac{6kq}{m}$
 (B) $\frac{\sqrt{3}kq}{m}$
 (C) $\frac{\sqrt{3}kq}{2m}$
 (D) $\frac{2\sqrt{3}kq}{m}$

13. A nucleus X of mass number 218 in free state decays to emit an α -particle. Kinetic energy of the α -particle emitted is 6.7 MeV. The Q-value of the α -decay nuclear reaction is (in MeV)
- (A) 0.125
 (B) 6.825
 (C) 6.8
 (D) 6.7

14. Symbolic representation of photodiode is

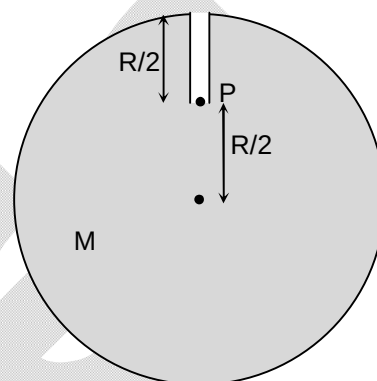


15. Two coherent sound sources in phase of frequencies of 256 Hz and 260 Hz are sounding together. If an observer records an intensity of I_0 at $t = 0$, then he will observe intensity at $t = \frac{1}{16}$ sec.

(A) I_0 (B) $\frac{I_0}{2}$
(C) $\frac{I_0}{4}$ (D) zero

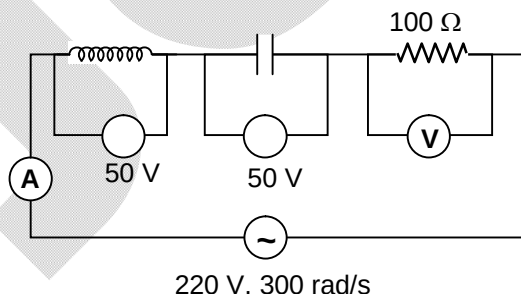
16. A particle is projected from a depth $\frac{R}{2}$ inside the earth such that it escapes to infinity. Mass of the particle is negligibly small is compared to mass of earth M . Then escape speed from the point P is

(A) $\sqrt{\frac{2GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$
(C) $\sqrt{\frac{13GM}{4R}}$ (D) $\sqrt{\frac{11GM}{4R}}$



17. For the AC circuit in the figure, choose the correct option having reading of AC voltmeter (V) and AC ammeter (A).

(A) 120 V, 1.2 A
(B) 220 V, 2.2 A
(C) 320 V, 3.2 A
(D) 220V, 1.2 A



18. The color code of a resistance is given in the figure. The value of resistance and tolerance, respectively are

(A) 47 kΩ, 10%
(B) 4.7 kΩ, 5%
(C) 470 Ω, 5%
(D) 470 kΩ, 5%

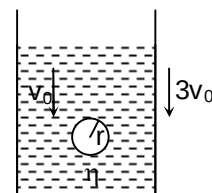


19. A non conducting ring of mass m and radius R a charge Q uniformly distributed over its circumference. The ring is placed on a rough horizontal surface such that the plane of the ring is parallel to the surface. A vertical magnetic field $B = B_0 t^2$ tesla is switched on at $t = 0$. After 2 second for switching on the magnetic field the ring is just about to rotate about vertical axis through its centre. m, R, Q, B are in SI unit. Then

(A) The induced electric field is steady with respect to time in the space
(B) the net electric force acting on the ring at any time t is $QB R t$
(C) Friction first acts at $t = 2$ sec

(D) The coefficient of static friction between the ring and the surface is $\frac{2B_0 R Q}{mg}$

20. A container filled with viscous liquid is moving vertically downward with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertical downwards in liquid has a speed v_0 with respect to ground. The coefficient of viscosity is η . There is no relative motion between the container and the liquid. Then at the shown instant, the power developed by viscous force with respect to ground frame is



- (A) $-6\pi\eta rv_0^2$ (B) zero
(C) $12\pi\eta rv_0^2$ (D) $-12\pi\eta rv_0^2$

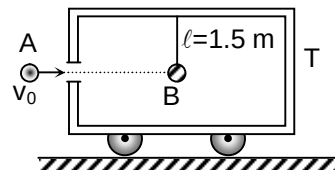
SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

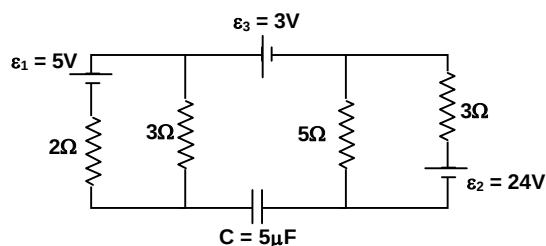
21. An unpolarized light of intensity 80 W/m^2 passes through three polarizing sheets whose pass axes make an angle of 30° , 60° and 30° with y-axis in same sense. The final transmitted intensity after passing through all three polarizers is $I \text{ W/m}^2$. Then find the value of $2I$.
22. A particle of specific charge $2 \times 10^{-2} \text{ C/kg}$ starts moving from the origin under the action of a uniform electric field $\vec{E} = (100 \text{ N/C})\hat{i}$ and magnetic field $\vec{B} = (5\text{T})\hat{k}$. Its velocity vector at $(a, b, 0) \text{ m}$ is $(4\hat{i} - 3\hat{j}) \text{ m/s}$. Then value of $4a$ is (in m)
23. An object is projected with a velocity of 10 m/s making an angle of 30° with horizontal from origin of x-y vertical plane having positive x-axis along horizontal and positive y-axis along the vertically upward direction. The equation of the trajectory is $y = Ax - Bx^2$ where x and y are co-ordinates at a point in meter. Find the value of $\sqrt{3} \left(\frac{A}{B} \right)$ (in SI unit). (Take $g = 10 \text{ m/s}^2$)
24. Each side of a cube is measured to be 7.203 m . If the total surface area of the cube considering the correct significant digits in m^2 is A , then find the value of $10A$.
25. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in 40 seconds is $\tau \text{ N-m}$. Find the value of $\pi\tau$. (Take $\pi^2 = 10$)

26. In the figure shown a ball A of mass $m = 1 \text{ kg}$ collide with another identical ball B with a speed $v_0 = 6 \text{ m/s}$ and sticks to it. The ball B is hanging inside a trolley T of mass $M = 4 \text{ kg}$. Find the maximum height (in cm) attained by the ball (A + B) from its initial level after the collision (Take $g = 10 \text{ m/s}^2$)



27. The solar constant (intensity of solar radiation) just at Venus surface measured to be 2600 W/m^2 . If the radius of the Sun is $7 \times 10^8 \text{ m}$, while the Venus-Sun distance is $1.1 \times 10^{11} \text{ m}$, then find the most probable wavelength in the solar radiation in nm. [Take $(1.81)^{\frac{1}{4}} = 1.16$, wein's constant $b = 2.9 \times 10^{-3} \text{ m-K}$] Sun can be assumed as a black body.

28. If energy stored in the capacitor C in the steady state is $U_0 \mu\text{J}$, then find $2U_0$.



29. A parallel plate capacitor with $A = 1.2 \text{ m}^2$ and separation $d = 1 \text{ mm}$ is charged by a current $i = (8A)e^{-\frac{t}{4\text{sec}}}$. Consider a plane surface X of area 0.6 m^2 parallel to the plates and drawn symmetrically between the plates. Find the displacement current (in mA) through the plane surface X at $t = 4 \text{ sec}$. (Take $e^{-1} = 0.37$)
30. A submarine A travelling at 18 km/hr is being chased along the line of its velocity by another submarine B travelling at 27 km/hr . B send a Sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν (in Hz) is close to (take speed of sound in water = 1500 m/s)

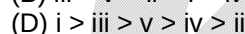
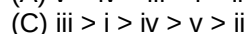
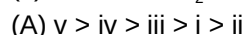
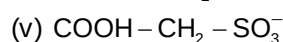
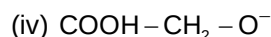
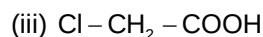
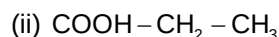
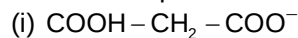
Chemistry

PART – B

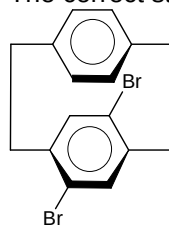
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Compare the acidic strength of following compounds and arrange in decreasing order. Choose the correct option



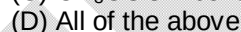
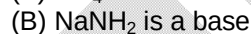
32. The correct statement about the following molecule is:



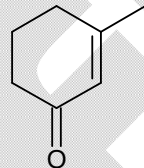
- (A) Molecule is chiral and does not possess chiral centre.
 (B) Molecule is chiral and possess chiral centre.
 (C) Molecule is achiral due to plane of symmetry.
 (D) Molecule is achiral due to axis of symmetry.

33. Choose the correct statement:

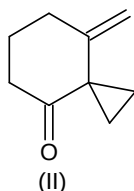
In liquid ammonia:



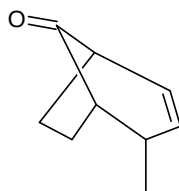
34. The product(s) are :



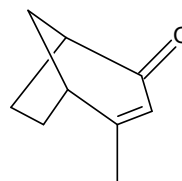
(I)



(II)



(III)



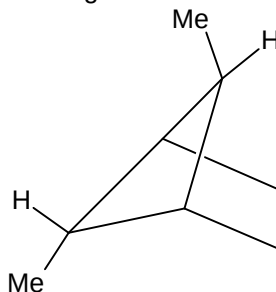
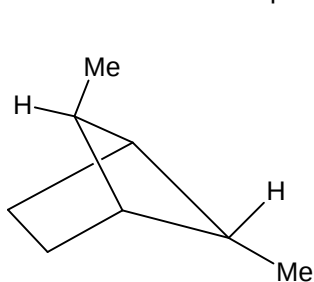
(IV)

- (A) I and III
 (C) I and II

- (B) II and IV
 (D) II and III

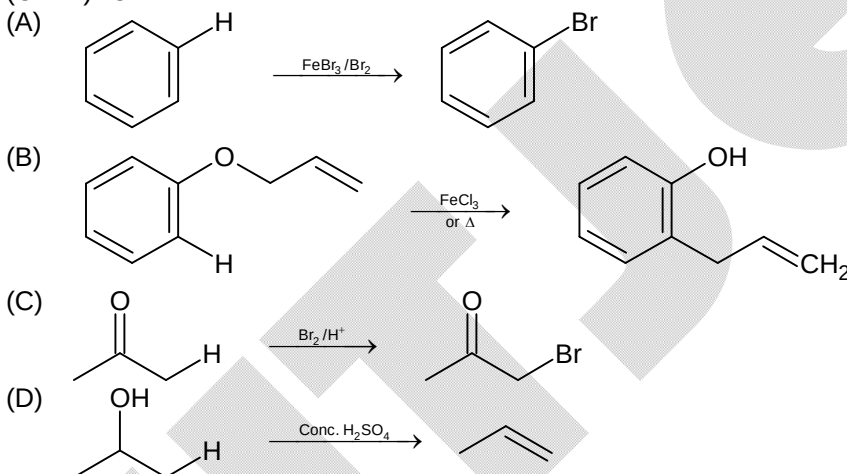
35. Considering N are central atom, the structure of $(\text{CH}_3)_2\text{NCS}$ and $(\text{SiH}_3)_2\text{NCS}$ respectively, are:
 (A) Linear and Bent (B) Bent and Bent
 (C) Linear and Linear (D) Bent and linear

36. The correct relationship between the following structures is that they are:



- (A) Enantiomers (B) Identical
 (C) Diastereomers (D) Constitutional isomers

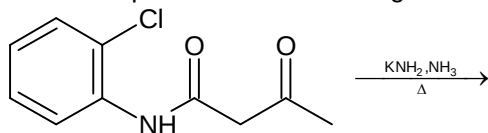
37. The reaction that is expected to show primary kinetic isotope effect for the indicated H-atom (C – H) is:



38. The pair in which both actinides show +3 oxidation state only is:
 (A) Ac and Lr (B) Ac and No
 (C) Cm and Bk (D) Cm and Lr
39. I_2 is violet in the solid as well as in gas phase. However, in acetone or ethanol, it turns brown. Choose the correct statement(s) for this colour change:
 (i) Dissociation of I_2 in atomic state.
 (ii) Interaction of low-lying σ^* orbital of iodine with lone pair of O (solvent).
 (iii) Formation of a charge transfer complex.
 (A) (i) only (B) (ii) only
 (C) (i) and (ii) only (D) (iii) only

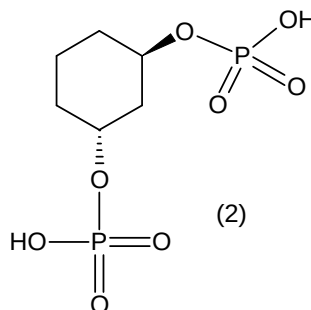
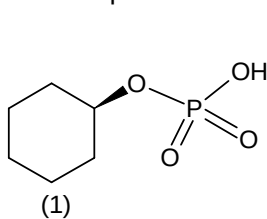
40. An atomic orbital is given by $nr^2 e^{-\frac{r}{3a_0}} (3\cos^2\theta - 1)$ in hydrogen atom. The orbital represents:
 (A) 2p (B) 3p
 (C) 3d (D) 4d

41. The magnitude of bond angles in gaseous NF_3 , SbF_3 and SbCl_3 follows the order:
 (A) $\text{NF}_3 > \text{NH}_3 > \text{SbCl}_3$ (B) $\text{SbCl}_3 > \text{NH}_3 > \text{NF}_3$
 (C) $\text{NH}_3 > \text{NF}_3 > \text{SbCl}_3$ (D) $\text{NF}_3 > \text{SbCl}_3 > \text{NH}_3$
42. Consider the following chemical reactions in gas phase
 Reaction I : $\text{H} + \text{D}_2 \longrightarrow \text{HD} + \text{D}$
 Reaction II : $\text{D} + \text{H}_2 \longrightarrow \text{HD} + \text{H}$
 Which of the following statement is true?
 (A) Reaction I and II, both are endothermic.
 (B) Reaction I is endothermic, Reaction II is exothermic.
 (C) Both reactions are thermo neutral as they involve isotopes which differ by the number of neutrons, which do not play any role in chemical reactions.
 (D) Reaction I and II both are exothermic.
43. Predict the product of the following reaction

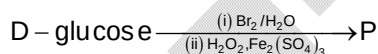


- (A)
- (B)
- (C)
- (D)

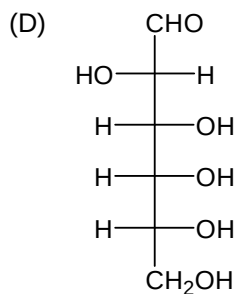
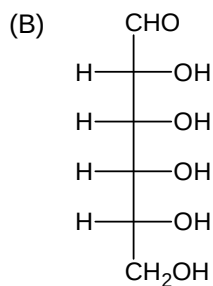
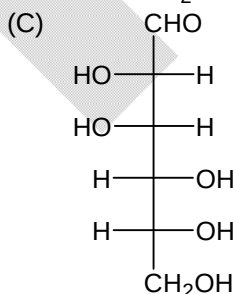
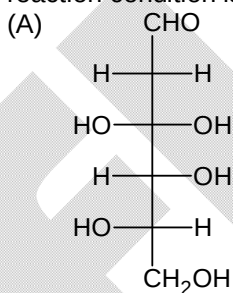
44. Compound 2 and compound 1 were mixed in a 1 : 4 molar ratio. The mixture was subjected to an ammonium molybdate test for phosphate estimation. The result of the molybdated test yielded a phosphate concentration of 36.0 mM for the mixture. What are the concentration of compound 1 and compound 2 in the mixture?



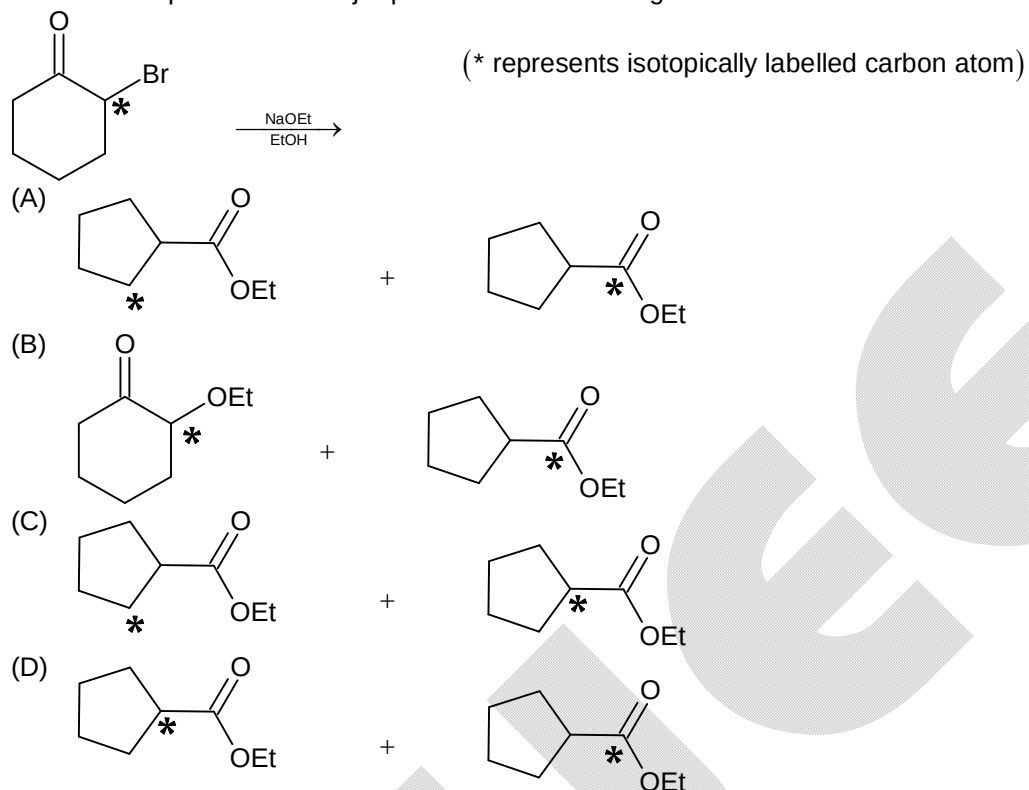
- (A) Compound 1 : 30 mM Compound 2 : 6 mM
 (B) Compound 1 : 24 mM Compound 2 : 6 mM
 (C) Compound 1 : 28.8 mM Compound 2 : 7.2 mM
 (D) Compound 1 : 24 mM Compound 2 : 12 mM
45. Which of the following can't be used for the given transformation?
- $$\text{H}_3\text{C}-\text{C}(=\text{O})-\text{C}(\text{CH}_3)_3 \longrightarrow \text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)_3$$
- (A) Clemmenson reduction (B) Wolf-Kishner reduction
 (C) Mozingo reduction (D) Bouveault-Blanc reduction
46. A mixture of acetone and CCl_4 can be separated by
- (A) Azeotropic distillation (B) Fractional distillation
 (C) Steam distillation (D) Vacuum distillation
47. In the following reaction of D-(+)glucose a product P is formed.



Amongst the following compounds, the one which will give the same product (P) under identical reaction condition is:



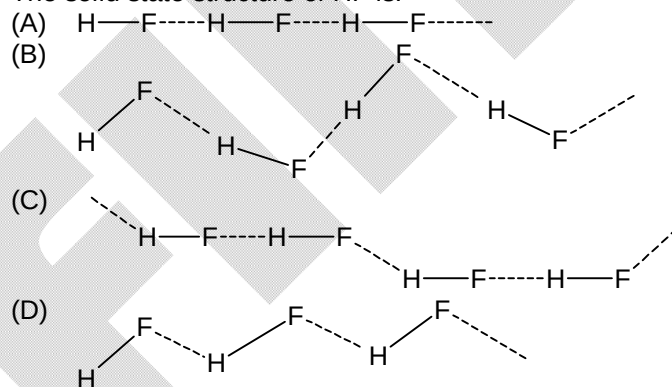
48. The correct option for the major product of the following reaction is:



49. The salt bridge in a Galvanic cell allows the flow of:

(A) ions but not electrons
 (B) both ions and electrons
 (C) electrons but not ions
 (D) neither ions nor electrons

50. The solid state structure of HF is:

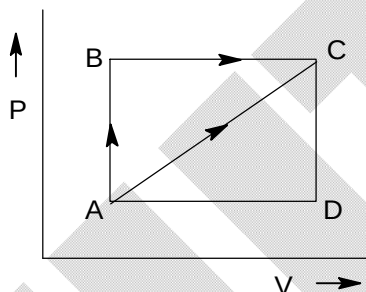


SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The number of stereoisomers for the complex $[\text{RhH}(\text{C} \equiv \text{CR})_2(\text{PMe}_3)_3]$ is:
52. The total number of lone pairs of electron on all atoms in cyanogens is 'x' and thiocyanogen is 'y'. Report $(x + y) = ?$
53. For the cell $\text{Cd} | \text{CdCl}_2 || \text{AgCl} | \text{Ag}$; $E^\circ_{\text{cell}} = 0.675 \text{ V}$ and $\left(\frac{dE^\circ}{dT}\right) = -6.5 \times 10^{-4} \text{ VK}^{-1}$ at 27°C . The $\Delta H^\circ (\text{kJ mol}^{-1})$ value for the cell is:
54. An organic compound is estimated through Dumas method and was found to evolve 264 g CO_2 , 72 g of H_2O and 28 g of N_2 . If the formula of the compound is $\text{C}_x\text{H}_y\text{N}_z$. Then $\left(\frac{x+y}{z}\right)$ is:
55. When glycinium hydrochloride is titrated against NaOH, pH at the first half equivalence point is 2.40 and the pH at the second half equivalence point is 9.60. The pH at first equivalence point is:
56. A thermodynamic process is shown below:
 Given: $P_A = 3 \times 10^4 \text{ Pa}$, $P_B = 8 \times 10^4 \text{ Pa}$, $V_A = 2 \times 10^{-3} \text{ m}^3$, $V_D = 5 \times 10^{-3} \text{ m}^3$. In the process AB, 600 J of heat is added and in BC 200 J of heat is added to the system. If the change in internal energy of the system in the process AC is x; then report $\left(\frac{x}{20}\right)$. x is calculated in Joule.



57. 10.6 g of benzaldehyde was reacted with an excess of acetophenone to produce 10.4 g of the enone product according to the given equation. Calculate the % yield of the product.
 $\text{Benzaldehyde} + \text{Acetophenone} \xrightarrow[2. \Delta]{1. \text{NaOEt}} \text{Product}$
58. Assume that the reaction of MeMgBr with ethyl acetate proceeds with 100% conversion to tert – butanol. The volume of 0.2 M solution of MeMgBr required to convert 10 mL of a 0.025 M solution of ethyl acetate is x. Report the value of $10x$ in mL.

59. The mechanism of enzyme catalysis is as follows:



The rate of product formation is given as:

$$\frac{+dP}{dt} = \frac{k_1 k_2 [E_0][S]}{k_{-1} + k_2 + k_1 [S]}, \text{ where } [E_0] \text{ is the total enzyme concentration, } [S] \text{ is substrate concentration.}$$

The rate of product formation when the substrate concentration is very large has a limiting value of 0.02 Ms^{-1} . At a substrate conc. of 250 mg/dm^3 , the rate is half this value. The value of $\frac{k_1}{k_{-1}}$ (in dm^3 / kg) assuming that $k_2 \ll k_{-1}$ is $x \times 10^3$. What is the value of x ?

60. The maximum concentration of equimolar solution of ferrous sulphate and potassium sulphide so that when mixed in equal volumes, there is no precipitate of iron sulphide is $C \times 10^{-9} \text{ M}$. Report the value of C . [Given: $K_{sp}(\text{FeS}) = 6.3 \times 10^{-18} \text{ M}^2$.

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If a non constant function $f(x) = \frac{2x^2 - kx + 5}{x^2 + 3x + 4}$ is bounded, then k can not have values is equal to
 (A) -2 (B) -6
 (C) 2 (D) 6
62. If $\sqrt{2} < \alpha, \beta < \sqrt{3}$, $\alpha \neq \beta$, then the value of $\left[\frac{\alpha - \beta}{\cot^{-1} \alpha - \cot^{-1} \beta} \right]$ is (where [.] denotes G.I.F.)
 (A) -4 (B) 4
 (C) 3 (D) -5
63. Let real number a satisfy $\{a\} + \left\{\frac{1}{a}\right\} = 1$ and b is the value of $\{a^3\} + \left\{\frac{1}{a^3}\right\}$ (where {.] denotes fractional part function), then the value of $\int_0^b \frac{x}{1+x} \sqrt{1-x^2} dx$ is
 (A) $2 + \frac{\pi}{4}$ (B) $2 - \frac{\pi}{4}$
 (C) $1 - \frac{\pi}{4}$ (D) $1 + \frac{\pi}{4}$
64. For a real number 'a' such that the equation $a(\sin x + \sin(2x)) = \sin 3x$ has more than one solution in the interval $(0, \pi)$ the set of all such value of a is
 (A) $(-3, -2)$ (B) $(-3, 1) - \{-2\}$
 (C) $(-2, 1)$ (D) $(0, 1)$
65. Let c be a real number and let z_1 and z_2 be the two complex number satisfying the equation $z^2 - cz + 10 = 0$, points $z_1, z_2, \frac{1}{z_1}$ and $\frac{1}{z_2}$ are vertices of (convex) quadrilateral in the complex plane, when the area is maximum then the value of c is
 (A) $4\sqrt{5}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{5}$ (D) $8\sqrt{5}$
66. If the equation $\frac{\alpha^2}{1 - \tan^2 x} = \frac{\sin^2 x + \alpha^2 - 2}{\cos 2x}$ has a solution, then
 (A) $\alpha^2 < 1$ (B) $\alpha^2 > 2$
 (C) $\alpha^2 \geq 1$ (D) $\alpha \in \mathbb{R}$
67. A polynomial of degree four with leading coefficient 1 and integer coefficients, has two zeros, both of which are integer. Which of the following can also be a zero of the polynomial ?
 (A) $\frac{1+i\sqrt{11}}{2}$ (B) $\frac{1+i}{2}$
 (C) $\frac{1}{2} + i$ (D) $1 + \frac{i}{2}$

68. Circle $x^2 + y^2 - 6x - 4y + 11 = 0$ is rolled up by 4 units along a tangent to it at the point (2, 3) let this be circle C_1 , C_2 is the mirror image of circle C_1 about the tangent. A and B are centres of circles C_1 and C_2 , C and D are the feet of perpendicular from A and B respectively upto x-axis, then the area of trapezium ABCD equals to
- (A) $(2 + 2\sqrt{2})$ (B) $(3 + 2\sqrt{2})$
 (C) $(6 + 4\sqrt{2})$ (D) $(4 + 4\sqrt{2})$
69. A point P is selected at random from the interior of the pentagon with vertices $A = (0, 2)$, $B = (4, 0)$, $C = (2\pi + 1, 0)$, $D = (2\pi + 1, 4)$ and $E = (0, 4)$. What is the probability that $\angle APB$ is obtuse ?
- (A) $1/5$ (B) $1/4$
 (C) $3/8$ (D) $5/16$
70. How many 15 letter arrangements of 5 A's, 5B's and 5 C's have no A's in the first 5 letter no B's in the next 5 letters and no C's in the last 5 letters
- (A) $\sum_{k=0}^5 ({}^5C_k)^3$ (B) $\sum_{k=0}^5 ({}^5P_k)^3$
 (C) 2^{15} (D) $2^5 \cdot 3^5$
71. The value of $\sum_{i=0}^n \sum_{j=1}^n {}^nC_j \cdot {}^jC_i \quad i \leq j$
- (A) 3^n (B) $3^n + 1$
 (C) $3^n - 1$ (D) 3^{n+1}
72. A parabola $y = ax^2 + bx + c$ is reflected about the x-axis, the parabola and its reflection are translated horizontally five units in opposite direction to become the graph of $y = f(x)$ and $y = g(x)$ respectively. Then which of the following describes the graph of $y = f(x) + g(x)$
- (A) A horizontal line (B) the graph of a cubic function
 (C) a non horizontal line (D) A parabola
73. If z_1 and z_2 are two complex number such that $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ where $z_1 = 4 + 6i$ and $z_2 = 3 + 5i$, then locus of z is
- (A) straight line with slope $-\frac{1}{2}$ (B) circle with radius $\frac{1}{\sqrt{2}}$
 (C) hyperbola with eccentricity $\sqrt{2}$ (D) hyperbola with eccentricity $\frac{5}{2}$
74. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$, then $\int_{1/10}^{9/10} f(f(x))dx$ is equal to
- (A) $\frac{4}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{5}$ (D) 0
75. If $I_1 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}} x^2 dx$ and $I_2 = \int_0^1 (1 - (1-x^3)^{\sqrt{2}})^{\sqrt{3}+1} x^2 dx$, then $\frac{I_1 - I_2}{I_2} = \frac{1}{k}$, then k is equal to
- (A) $(\sqrt{3} + 1)\sqrt{2}$ (B) $(\sqrt{3} - 1)\sqrt{2}$
 (C) $(\sqrt{3} + \sqrt{2})$ (D) $3(\sqrt{6} + \sqrt{2})$

76. Total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^{10} + 1}{t^{10} + t^4 + 1} dt = 6x - 5$ is
 (A) 0 (B) 1
 (C) 2 (D) 3
77. A continuous and differentiable function $f(x)$ satisfy the functional equation $f(x + y) = \frac{f(x)f(y)}{f(x) + f(y)}$,
 $f(1) = 32$ then $f(4) =$
 (A) 8 (B) 6
 (C) 4 (D) 2
78. $\lim_{x \rightarrow \infty} |1 - a^4|^x \sin \frac{5}{|1 - a^2|^x} a^2 > 2$ is
 (A) 0 (B) 5
 (C) ∞ (D) does not exist
79. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2 - x)$ and $f''(x) < 0 \forall x \in (0, 2)$ then which is true ?
 (A) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $(0, 2)$
 (B) $g(x)$ increases in $(0, 2)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (C) $g(x)$ increases in $\left(\frac{4}{3}, 2\right)$ and decreases in $\left(0, \frac{4}{3}\right)$
 (D) $g(x)$ increases in $\left(0, \frac{4}{3}\right)$ and decreases in $\left(\frac{4}{3}, 2\right)$
80. Area bounded by the curve $y = 1$ and $y = \frac{\sin x + \cos x + |\sin x - \cos x|}{2}$ for $x \in [0, 2\pi]$ is
 (A) $2\pi - \sqrt{2}$ (B) $2\pi - 2\sqrt{2}$
 (C) $2(\pi - \sqrt{2} - 1)$ (D) $2(\pi + \sqrt{2} - 3)$

SECTION - B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = \sqrt{ax^2 + bx}$, for how many real values of a is there at least one positive value of b for which the domain of $f(x)$ and the range of $f(x)$ are the same set
82. Lines $x + y + 4 = 0$, $2x + ky + 2 = 0$ intersects the circle $x^2 + y^2 + 2x + 4y + 1 = 0$ and $x^2 + y^2 + 4x + 6y + 4 = 0$ as 4 points which are concyclic, then the value of k is
83. The number of distinct values of determinant of order 2×2 whose entries are from the set $\{-1, 0, 1\}$ is
84. Krishna writes the number 1, 2, 3, ..., 9 on separate cards, one number per card, he wishes to divide the cards into 3 groups of 3 cards so that the sum of the numbers in each group will be the same. In how many ways can this be done ?

85. Fifteen integers $a_1, a_2, a_3, \dots, a_{15}$ are arranged in order on a number line. The integers are equally spaced and have the property that $1 \leq a_1 \leq 10$, $13 \leq a_2 \leq 20$ and $241 \leq a_{15} \leq 250$. What is the sum of digits of a_{14} ?
86. In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term is 4000. What is the first term
87. A data set consists of 6 (not all distinct) positive integers : 1, 7, 5, 2, 5 and x, the Arithmetic Mean of the 6 numbers equals a value in the data set. What is the sum of all possible values of x
88. The area bounded by the graph of $y^2 + 2xy + 40|x| = 400$ is
89. Let the distance between two point (x_1, y_1) and (x_2, y_2) is given by $|x_1 - x_2| + |y_1 - y_2|$ for how many points P with integer coordinates is the distance between P and the origin less than or equal to 20
90. Two integers are inserted into list 3, 3, 3, 8, 11, 12, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of the two additional numbers